

TURBULENT BL

06/02/24

$$\bar{u} \frac{\partial \bar{u}}{\partial x} + \bar{v} \frac{\partial \bar{u}}{\partial y} = -\frac{1}{\rho} \frac{\partial \bar{P}}{\partial x} + \nu \frac{\partial^2 \bar{u}}{\partial y^2} - \frac{\partial}{\partial y} (\overline{u'v'})$$

$$\bar{u} \frac{\partial \bar{T}}{\partial x} + \bar{v} \frac{\partial \bar{T}}{\partial y} = \alpha \frac{\partial^2 \bar{T}}{\partial y^2} - \frac{\partial}{\partial y} (\overline{v'T'})$$

$$-\rho \overline{u'v'} = \rho \epsilon_m \frac{\partial \bar{u}}{\partial y} \quad : \text{Reynold's Stress / Eddy Shear Stress.}$$

$$-\rho c_p \overline{v'T'} = \rho c_p \epsilon_h \frac{\partial \bar{T}}{\partial y} \quad : \text{Eddy Heat flux.}$$

$$\left[\begin{array}{l} \epsilon_m: \text{Eddy momentum diffusivity} \\ \epsilon_h: \text{Eddy heat diffusivity} \end{array} \right]$$

Replacing these terms,

$$\bar{u} \frac{\partial \bar{u}}{\partial x} + \bar{v} \frac{\partial \bar{u}}{\partial y} = -\frac{1}{\rho} \frac{\partial \bar{P}}{\partial x} + \frac{1}{\rho} \frac{\partial}{\partial y} \left[\mu \frac{\partial \bar{u}}{\partial y} - \rho \overline{u'v'} \right]$$

$$\bar{u} \frac{\partial \bar{T}}{\partial x} + \bar{v} \frac{\partial \bar{T}}{\partial y} = \frac{1}{\rho c_p} \frac{\partial}{\partial y} \left[k \frac{\partial \bar{T}}{\partial y} - \rho c_p \overline{v'T'} \right]$$

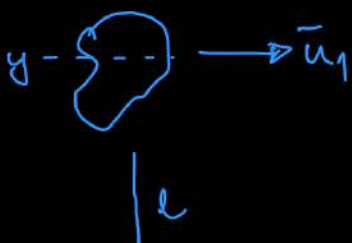
$$\bar{u} \frac{\partial \bar{u}}{\partial x} + \bar{v} \frac{\partial \bar{u}}{\partial y} = -\frac{1}{\rho} \frac{\partial \bar{P}}{\partial x} + \frac{1}{\rho} \frac{\partial}{\partial y} \left[\mu \frac{\partial \bar{u}}{\partial y} + \rho \epsilon_m \frac{\partial \bar{u}}{\partial y} \right]$$

$$\bar{u} \frac{\partial \bar{T}}{\partial x} + \bar{v} \frac{\partial \bar{T}}{\partial y} = \frac{1}{\rho c_p} \frac{\partial}{\partial y} \left[k \frac{\partial \bar{T}}{\partial y} + \rho c_p \epsilon_h \frac{\partial \bar{T}}{\partial y} \right]$$

$$\bar{u} \frac{\partial \bar{u}}{\partial x} + \bar{v} \frac{\partial \bar{u}}{\partial y} = -\frac{1}{\rho} \frac{\partial \bar{P}}{\partial x} + \frac{\partial}{\partial y} \left[(\nu + \epsilon_m) \frac{\partial \bar{u}}{\partial y} \right] \quad \text{--- (i)}$$

$$\bar{u} \frac{\partial \bar{T}}{\partial x} + \bar{v} \frac{\partial \bar{T}}{\partial y} = \frac{\partial}{\partial y} \left[(\alpha + \epsilon_h) \frac{\partial \bar{T}}{\partial y} \right] \quad \text{--- (ii)}$$

MIXING LENGTH MODEL



Assumptions:

- > l is a linear function of y
- > Transverse movement does not change identity of fluid

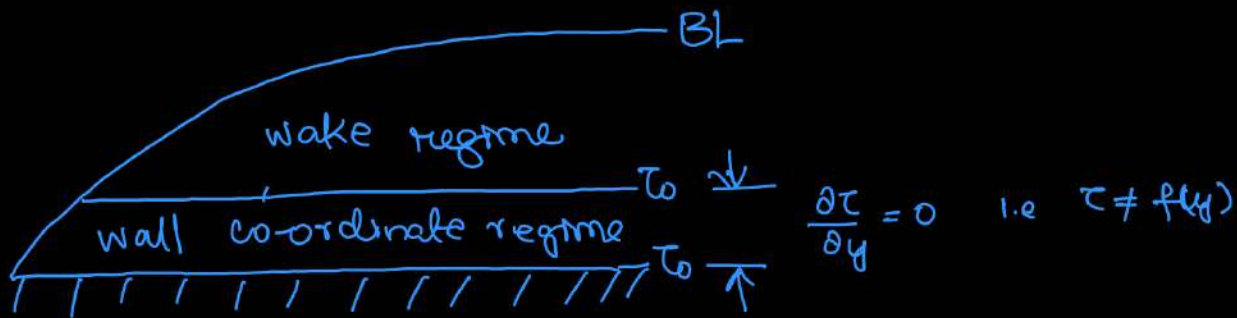


packet $\Rightarrow \Delta U(u) \approx u$
 $\rightarrow$ Both fluctuating comp have same order

$$O(\bar{u}(y)) - O(\bar{u}(y-l)) \approx u' \quad \text{--- 2nd assumption}$$

$$\approx l \frac{\partial \bar{u}}{\partial y}$$

$l = \kappa y$ where κ is Von-Karman's constant



Within wall co-ordinate regime, inertia $= 0$; $\frac{\partial \bar{P}}{\partial x} = 0$

$$u^+ = \frac{\bar{u}}{u^*} \quad \text{where } u^* = \text{frictional velocity}$$

$$= \left(\frac{\tau_0}{\rho} \right)^{1/2}$$

$$v^+ = \frac{\bar{v}}{u^*} \quad y^+ = y \left(\frac{u^*}{\nu} \right) \quad x^+ = x \left(\frac{u^*}{\nu} \right)$$

$$O(v') = l \frac{\partial \bar{u}}{\partial y} = O(u') \quad \text{--- 3rd assumption}$$

$$O(\overline{u'v'}) = l^2 \left(\frac{\partial \bar{u}}{\partial y} \right)^2 \Rightarrow \epsilon_M = l^2 \left| \frac{\partial \bar{u}}{\partial y} \right|$$

Eqn (1) gives,

$$\frac{\partial}{\partial y} \left[(\nu + \epsilon_M) \frac{\partial \bar{u}}{\partial y} \right] = 0$$

$$\boxed{(\nu + \epsilon_M) \frac{\partial \bar{u}}{\partial y} = \text{const} = \frac{\tau_0}{\rho}}$$

Replacing $\bar{u}$ in terms of u^+ ,

$$(\nu + \epsilon_M) \frac{u^{*2}}{\nu} \frac{\partial u^+}{\partial y^+} = \frac{\tau_0}{\rho}$$

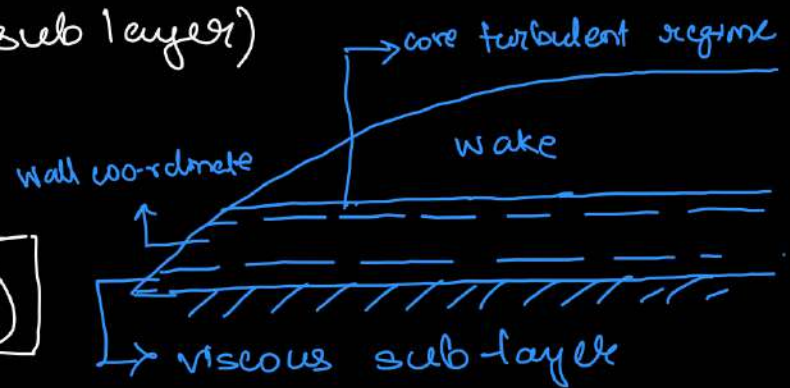
$$\boxed{\left[1 + \frac{\epsilon_M}{\nu} \right] \frac{\partial u^+}{\partial y^+} = 1}$$

when $\nu \gg \epsilon_m$, (very close to the wall.)

$$\frac{\partial u^+}{\partial y^+} = 1 \quad (\text{viscous sub layer})$$

$$\int_0^{y^+} \partial u^+ = \int_0^{y^+} \partial y^+$$

$$u^+ = y^+ \quad (0 < y^+ < y_{vsl}^+)$$



when $\epsilon_m \gg \nu$,

(core turbulent regime)

$$\frac{\epsilon_m}{\nu} \frac{\partial u^+}{\partial y^+} = 1$$

$$\frac{\nu^2}{\nu} \left| \frac{\partial \bar{u}}{\partial y} \right| \frac{\partial u^+}{\partial y^+} = 1 \Rightarrow \frac{(\kappa y)^2}{\nu} \left| \frac{\partial \bar{u}}{\partial y} \right| \frac{\partial u^+}{\partial y^+} = 1$$

$$\frac{\kappa^2}{\nu} \left| \frac{\partial \bar{u}}{\partial y} \right| \frac{y^{+2} \nu^2}{u^{+2}} \frac{\partial u^+}{\partial y^+} = 1$$

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$$\frac{\kappa^2}{\nu} \frac{y^{+2} \nu^2}{u^{+2}} \left| \frac{\partial \bar{u}}{\partial y} \right| \frac{\partial u^+}{\partial y^+} = 1$$

$$\kappa^2 \left| \frac{\partial \bar{u}}{\partial y} \right| \frac{\partial u^+}{\partial y^+} \cdot y^{+2} = 1$$

$$y^+ \frac{\partial u^+}{\partial y^+} = \frac{1}{\kappa} \Rightarrow \kappa \partial u^+ = \frac{\partial y^+}{y^+}$$

$$\int_{u_{vsl}^+}^{u^+} \kappa \partial u^+ = \int_{y_{vsl}^+}^{y^+} \frac{\partial y^+}{y^+}$$

$$\kappa [u^+ - u_{vsl}^+] = \ln y^+ - \ln y_{vsl}^+$$

$$u^+ = \frac{1}{\kappa} [\ln y^+] - \frac{1}{\kappa} \ln y_{vsl}^+ + y_{vsl}^+$$

$$= A \ln y^+ + B \quad \text{form}$$

+ print in the wake region. leads to

Assume, $u^1 = f(y^1)$ in the work of
Ponditi 11th theory (?).

VELOCITY PROFILE IN CTR/WC

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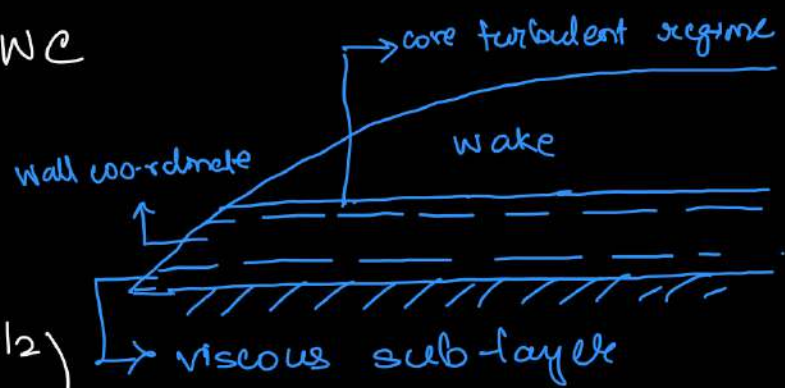
$u^+ = f(y^+)$ what is f ?

defined only for VSL & WC

$$\frac{\bar{u}}{u^*} = f\left(\frac{\bar{y}}{\nu} \left(\frac{u^*}{\nu}\right)\right)$$

At edge of BL,

$$\frac{u_\infty}{(\tau_0/\rho)^{1/2}} = f\left(\delta \cdot \frac{(\tau_0/\rho)^{1/2}}{\nu}\right) \quad \text{--- (i)}$$



In integral approach,

$$\frac{d}{dx} \int_0^\delta u(u_\infty - u) dy = -\frac{1}{\rho} \frac{d\tau_0}{dx} + \frac{d u_\infty}{dx} \int_0^\delta u dy + \nu \left(\frac{\partial u}{\partial y}\right)_{y=0}$$

$$\frac{d}{dx} \int_0^\delta u(u_\infty - u) dy = \left(\frac{\tau_0}{\rho}\right) \quad \text{--- (ii)}$$

Most popular velocity profile in BL is

Poandtl's 1/7th power law.

$$u^+ = 8.7(y^+)^{1/7}$$

$$\frac{\tau_0}{\rho U_\infty^2} = 0.0225 \left(\frac{U_\infty \delta}{\nu}\right)^{-1/4} \quad (7.57)$$

$$\frac{\delta}{x} = 0.37 \left(\frac{U_\infty x}{\nu}\right)^{-1/5} \quad (7.58)$$

$$\frac{\tau_0}{\rho U_\infty^2} = \frac{1}{2} C_{f,x} = 0.0296 \left(\frac{U_\infty x}{\nu}\right)^{-1/5} \quad (7.60)$$

$$\frac{\tau_{0-x}}{\rho U_\infty^2} = \frac{1}{2} C_{f,0-x} = 0.037 \left(\frac{U_\infty x}{\nu}\right)^{-1/5} \quad (7.60')$$

TEMPERATURE PROFILE - BL

$$-q''_{app} = k \frac{\partial \bar{T}}{\partial y} - \rho c_p \bar{v}'T' = \rho c_p (\alpha + \epsilon_H) \frac{\partial \bar{T}}{\partial y}$$

$$u \frac{\partial \bar{T}}{\partial x} + \bar{v} \frac{\partial \bar{T}}{\partial y} = \frac{\partial}{\partial y} \left[(\alpha + \epsilon_H) \frac{\partial \bar{T}}{\partial y} \right]$$

Assuming q''_0 is constant and there is no bulk energy transport.

$$\frac{\partial}{\partial y} \left[(\alpha + \epsilon_H) \frac{\partial \bar{T}}{\partial y} \right] = 0$$

$$(\alpha + \epsilon_H) \frac{\partial \bar{T}}{\partial y} = \text{const} = \frac{q''_0}{\alpha_p}$$

$$\frac{\alpha_p}{q''_0}$$

$$= \frac{\partial \bar{y}}{(\alpha + \epsilon_H)}$$

$$\frac{\alpha_p u^+}{q''_0} \partial(T_0 - \bar{T}) = \frac{\partial y^+}{\left(\frac{\alpha}{\nu} + \epsilon_H/\nu\right)}$$

$$\frac{\alpha_p u^+}{q''_0} \partial(T_0 - \bar{T}) = \frac{\partial y^+}{\left(\frac{1}{Pr} + \left(\frac{\epsilon_H}{\epsilon_M}\right)\left(\frac{\epsilon_M}{\nu}\right)\right)} = \left(\frac{1}{Pr} + \frac{1}{Pr_T} \frac{\epsilon_M}{\nu}\right)$$

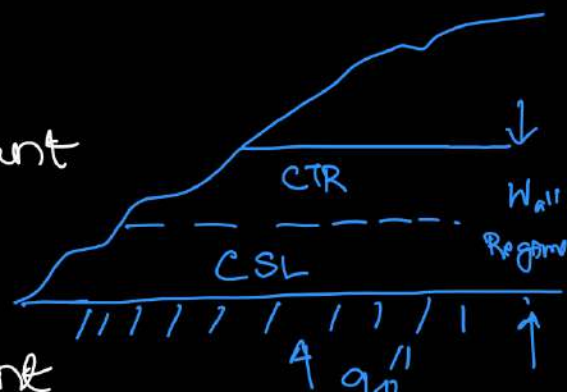
$$T^+(x^+, y^+) = \frac{\alpha_p u^+}{q''_0} (T_0 - \bar{T}) \quad \left[\text{non dimensionalizing temperature} \right]$$

$$\frac{\partial T^+}{\partial y^+} = \frac{1}{\left[\frac{1}{Pr} + \frac{1}{Pr_T} \left(\frac{\epsilon_M}{\nu}\right)\right]}$$

Two regimes,

(a) where conduction is dominant
(conduction sub layer)

(b) Turbulent region, where



eddy diffusivity is

$$T^+ = \int_0^{CSL} \frac{dy^+}{\left(\frac{1}{Pr} + \right)} + \int_{CSL}^{\delta_T} \frac{dy^+}{\left(\frac{1}{Pr} + \frac{1}{Pr_T} \left(\frac{y^+}{\delta_T}\right)\right)}$$

At CSL, $\boxed{T^+ = Pr y^+} \quad 0 \leq y^+ \leq y_{CSL}^+$

Derivation from Bejan [Pg 339]

$$\underbrace{\bar{u} \frac{\partial \bar{T}}{\partial x}}_{\text{Bulk}} + \underbrace{\bar{v} \frac{\partial \bar{T}}{\partial y}}_{\text{conduction diffusion}} = \underbrace{\frac{\partial}{\partial y} \left[(\alpha + \epsilon_H) \frac{\partial \bar{T}}{\partial y} \right]}_{\text{eddy diffusion}}$$

Ignoring bulk energy transport close to the wall,

$$\frac{\partial}{\partial y} \left[(\alpha + \epsilon_H) \frac{\partial \bar{T}}{\partial y} \right] = 0$$

$$(\alpha + \epsilon_H) \frac{\partial \bar{T}}{\partial y} = \left[(\alpha + \epsilon_H) \frac{\partial \bar{T}}{\partial y} \right]_{@y=0} = \text{const}$$

$$q''_0 = - \rho c_p \left[(\alpha + \epsilon_H) \frac{\partial \bar{T}}{\partial y} \right]_{@y=0} \quad \text{wall heat flux}$$

$$\left[\alpha = \frac{k}{\rho c_p} \right]$$

$$(\alpha + \epsilon_H) \frac{\partial \bar{T}}{\partial y} = \frac{-q''_0}{\rho c_p} \quad \left[\text{applicable sufficiently close to the wall} \right]$$

Introducing the wall coordinates y^+ ,

where $y^+ = \bar{y} \left(\frac{u^*}{\nu} \right)$

$$\left(\frac{\rho c_p}{-q''_0} \right) \frac{\partial \bar{T}}{\partial y^+} \left(\frac{u^*}{\nu} \right) = \frac{1}{(\alpha + \epsilon_H)}$$

$$\left(\rho c_p u^* \right) \frac{\partial (\bar{T} - T_0)}{\partial y^+} = \frac{1}{\left(\frac{1}{Pr} + \frac{1}{Pr_T} \left(\frac{y^+}{\delta_T} \right) \right)}$$

$$(-q''_0) \partial y^+ \quad \left(\frac{\alpha}{\nu} + \frac{\epsilon_H}{\nu} \right)$$

Defining $T^+(x^+, y^+)$ wall co-ordinate temp,

$$T^+(x^+, y^+) = (T_0 - \bar{T}) \frac{\rho c_p u^+}{q''_0}$$

$$\frac{\partial T^+}{\partial y^+} = \frac{1}{\left(\frac{\alpha}{\nu} + \frac{\epsilon_H}{\nu} \right)} = \frac{1}{1/Pr + \left(\frac{\epsilon_H}{\epsilon_M} \right) \left(\frac{\epsilon_M}{\nu} \right)}$$

$$\int_0^{T^+} dT^+ = \int_0^{y^+} \frac{dy^+}{\left(1/Pr + 1/Pr_T \left(\frac{\epsilon_M}{\nu} \right) \right)}$$

For the region $q''_0 = q''_{app} = \text{const}$ (close to the wall)
the temp profile is governed by
the Pr , Pr_T , ϵ_M/ν only

→ In the fully turbulent region of the
const T_{app} layer, $\frac{\epsilon_M}{\nu} = \frac{dy^+}{du^+} = Ky^+$

→ $\frac{1}{Pr_T} \left(\frac{\epsilon_M}{\nu} \right)$ therefore increases steadily
with y^+

→ Close to wall (near VSL) $\frac{\epsilon_M}{\nu} \rightarrow 0$

Close to wake region (near CTR) $\frac{\epsilon_M}{\nu} \rightarrow \alpha$
[large]

$$T^+ = \int_0^{y^+_{CSL}} \frac{dy^+}{\frac{1}{Pr}} + \int_{y^+_{CSL}}^{y^+} \frac{dy^+}{\frac{1}{Pr_T} \left(\frac{\epsilon_M}{\nu} \right)}$$

y^+_{CSL} : conduction sub layer thickness
[similar to viscous sub layer]

CSL: Molecular mechanism outweighs the eddy transport of heat

$$\overline{T^+} = \begin{cases} Pr_T y^+ & y^+ < y_{CSL}^+ \\ Pr_T y_{CSL}^+ + \frac{Pr_T}{K} \ln \frac{y^+}{y_{CSL}^+} & y^+ > y_{CSL}^+ \end{cases}$$

$y_{CSL}^+ \approx 13.2$
 $K \approx 0.91$
 $Pr_T \approx 0.9$
 $0.5 < Pr_T < 5$

1ST TERM 2ND TERM

As q_w'' is unknown, assumption that the eqn is valid till edge of BL yields,

$$\left(\frac{e c_p u^*}{-q_w''} \right) [\bar{T} - T_0] \Big|_{T_0} = Pr_T y_{CSL}^+ + \frac{Pr_T}{K} \ln \frac{\bar{y} \left(\frac{u^*}{\nu} \right)}{y_{CSL}^+} \Big|_{\delta}$$

$$e c_p u^* \left[\frac{T_0 - T_{\infty}}{q_w''} \right] = Pr_T y_{CSL}^+ + \frac{Pr_T}{K} \ln \left(\frac{\delta u^*}{\nu y_{CSL}^+} \right)$$

$$h = \left(\frac{q_w''}{T_0 - T_{\infty}} \right) \text{ and } \delta \text{ is unknown}$$

At the edge of the BL, according to the law of the wall, not Prandtl's $1/7$ th power law.

$$\frac{U_{\infty}}{u_*} = \frac{1}{K} \ln \left(\frac{\delta u^*}{\nu} \right) + B \quad \left[u^* = U_{\infty} \left(\frac{C_{f,x}}{2} \right)^{1/2} \right]$$

(u^*) (y^+) $\left[\because C_{f,x} = \frac{\tau_0}{\frac{1}{2} \rho U_{\infty}^2} \right]$

Substituting δ ,

$$\frac{e c_p u^*}{h} = Pr_T y_{CSL}^+ + \frac{Pr_T}{K} \left[\left(\frac{U_{\infty}}{u^*} - B \right) K - \ln y_{CSL}^+ \right]$$

$$\frac{h}{e c_p} = \frac{u^*}{Pr_T y_{CSL}^+ - \frac{Pr_T}{K} \ln y_{CSL}^+ - B Pr_T + \frac{Pr_T U_{\infty}}{u^*}}$$

$$\frac{h}{\rho c_p u_\infty} = \frac{(C_{f,x}/2)}{Pr_T + \left(\frac{C_{f,x}}{2}\right)^{1/2} \left[Pr_T y_{csl}^+ - B Pr_T - \frac{Pr_T \ln y_{csl}^+}{K} \right]}$$

(local Stanton no)

$$St_x = \frac{h}{\rho c_p u_\infty} = \frac{Nu_x}{Re_x Pr} = \frac{Nu_x}{Re_x Pr}$$

$St_x \cdot Pr^{2/3} = \frac{1}{2} C_{f,x}$

(Colburn Empirical)

Von-Karman solved the same problem but by considering 3 different layers for the integration.

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→ After fitting the T^+ curve to experimental data,

$T^+ = 2.195 \ln y^+ + 13.2 Pr - 5.66$

$$\left[y_{csl}^+ \cong 13.2 \quad Pr_T \cong 0.9 \quad K \cong 0.91 \quad B \cong 5.5 \right]$$

→ With same values, local St no 528 e

$St_x = \frac{\frac{1}{2} C_{f,x}}{0.9 + \frac{1}{2} (C_{f,x})^{1/2} [13.2 Pr - 10.25]}$

→ In the whole expression, only $C_{f,x}$ is a $f(Re)$.

→ $C_{f,x}$ however is a weak correlation of Re

→ Denominator has $Pr \sim O(1)$

$$\therefore St_x = K \cdot \frac{C_{f,x}}{2}$$

↳ $f(Pr)$

$K \sim Pr^{-2/3}$

[Prandtl's eqn]

$$St_n \cdot Pr^{2/3} = \frac{1}{2} C_{f,x} \quad \text{[Colburn's eqn]}$$

Von Karman 3 Layer Theory

$$St_n = \frac{\frac{1}{2} C_{f,x}}{1 + 5 \left(\frac{1}{2} C_{f,x} \right)^{1/2} \left[Pr - 1.7 \ln \left\{ 1 + \frac{5}{6} (Pr - 1) \right\} \right]}$$

$$\text{For } Pr = 1, \quad St_n = \frac{\frac{1}{2} C_{f,x}}{1 + 5 \left(\frac{1}{2} C_{f,x} \right)^{1/2}}$$

Reynolds Analogy

$$\frac{\tau_0}{\rho} = (\nu + \epsilon_m) \frac{d\bar{u}}{dy}$$

$$\tau_0 = (\mu + \rho \epsilon_m) \frac{d\bar{u}}{dy} \quad \text{--- (i)}$$

$$-\frac{q_0''}{\rho c_p} = (\alpha + \epsilon_h) \frac{d\bar{T}}{dy}$$

$$-q_0'' = (k + \rho c_p \epsilon_h) \frac{d\bar{T}}{dy} \quad \text{--- (ii)}$$

Dividing (i) by (ii),

$$\frac{\tau_0}{-q_0''} = \frac{(\mu + \rho \epsilon_m)}{(k + \rho c_p \epsilon_h)} \frac{d\bar{u}}{d\bar{T}}$$

$$\text{For } Pr = 1, Pr_T = 1, \alpha = \nu \Rightarrow \frac{k}{\rho c_p} = \frac{\mu}{\rho} ; \epsilon_m = \epsilon_h$$

$$\frac{\tau_0}{-q_0''} = \frac{(\mu + \cancel{\rho \epsilon_m})}{c_p (\mu + \cancel{\rho \epsilon_h})} \frac{d\bar{u}}{d\bar{T}}$$

$$\frac{d\bar{T}}{-q_0''} = \frac{d\bar{u}}{c_p \tau_0}$$

$$\frac{T_0}{-q_0''} = \frac{u_\infty}{C_p(T_\infty - T_0)}$$

$$[T_0 = \frac{1}{2} C u_\infty^2 C_{f,x}]$$

$$[q_0'' = h(\Delta T)]$$

$$\frac{1}{2} \frac{C u_\infty^2 C_{f,x}}{h \Delta T} = \frac{1}{C_p} \frac{u_\infty}{\Delta T}$$

$$\frac{h}{C_p u_\infty} = \frac{1}{2} C_{f,x}$$

$$St_x = \frac{1}{2} C_{f,x}$$

[Reynold's Analogy]

$Pr \sim f(\text{system property})$ [only if properties are const]

$Pr_\gamma \sim f(\text{flow property})$

If flow increased Pr const, Pr_γ changes.

Problem 3

To examine the wall shear stress on a flat plate, maintained at a constant temperature of 30°C ($T_w = 30^\circ\text{C}$). An air stream of uniform velocity ($u_\infty = 10\text{ m/s}$) and temperature ($T_\infty = 10^\circ\text{C}$) at a distance from the wing is blown over it. Flow is assumed to be turbulent from $x = 0$. At distance $x (= 1\text{ m})$ from the plate leading edge in the direction of the flow, the wall shear stress τ_w equal to 0.23 N m^{-2} is measured. In the same location, a wire of diameter $1\text{ }\mu\text{m}$ is placed at 0.19 mm above the wall to measure the fluid mean temperature, which is found to be equal to 25°C .

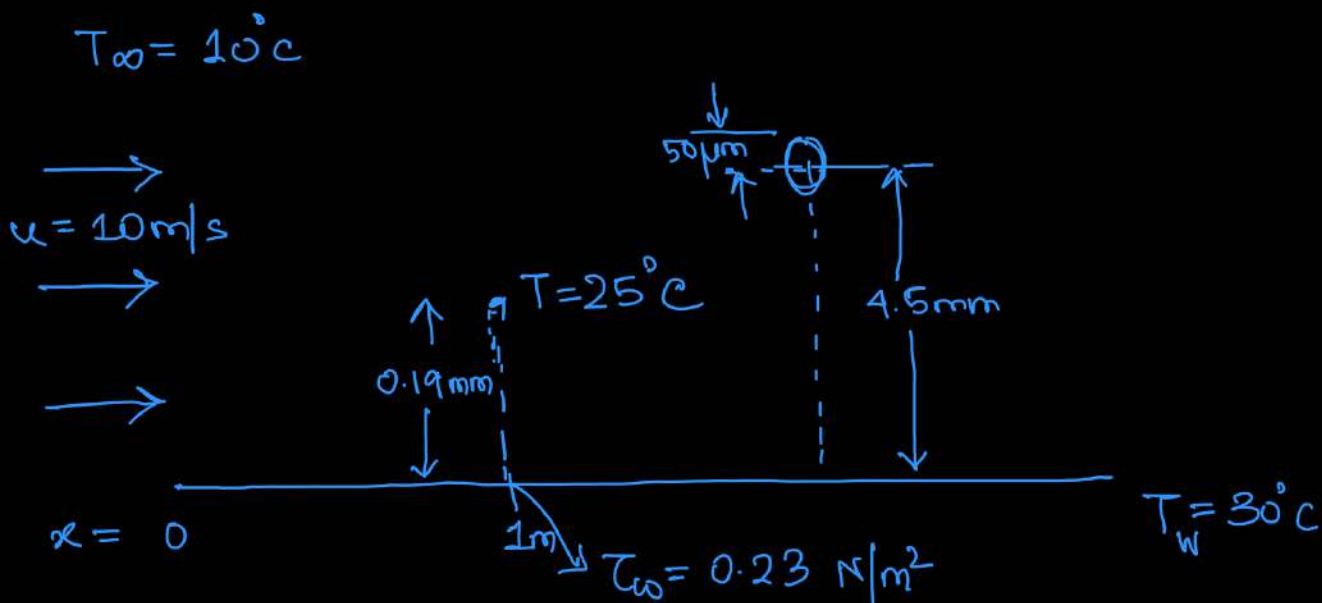
- Find the wall heat flux at the location of the measurement.
- Compare with the Colburn correlation and find the % difference relative to the direct measurement.
- Now the temperature measuring wire is replaced by a thermocouple. The thermocouple sensor has a spherical-shaped diameter $d \approx 50\text{ }\mu\text{m}$ and is placed at 4.5 mm above the wall. What is the expected temperature?
- What is the sensor size-induced uncertainty in the mean temperature measurement at this distance from the wall?
- The uniform velocity of air is now changed 50 m/s . Do you expect any change in the measured temperature? (For both measured temperature, initially by the wire located 0.19 mm above the wall and then by the thermocouple located 4.5 mm above the wall) Explain your answer with proper justification.

Assume the wall co-ordinate is valid till $y^+ < 200$

Physical properties of air:

$$\rho = 1.13\text{ kg m}^{-3}; \nu = 16.7 \times 10^{-6}\text{ m}^2\text{ s}^{-1}; k = 0.026\text{ W m}^{-1}\text{ K}^{-1}; Pr = 0.7; C_p = 10^3\text{ J kg}^{-1}\text{ K}^{-1}$$

The empirical constants are given as $Pr_t \approx 0.9$; $\kappa \approx 0.41$ and $y_{\text{cut}}^+ \approx 13.2$ for the above mention conditions



- q_w'' at $x=1\text{ m}$?
- Compare with Colburn's analogy
- Thermocouple @ 4.5 mm and $d = 50\text{ }\mu\text{m}$ of bead

SOLN:

(a) $y = 0.19 \text{ mm}$ $y^+ = y \left(\frac{u^+}{\nu} \right) = y \left(\frac{(\tau_0/\rho)^{1/2}}{\nu} \right)$

$y_{csL}^+ = 13.2$ (given)

$$y^+ = 0.19 \times 10^{-3} \times \left[\frac{(0.23/1.13)^{1/2}}{16.7 \times 10^{-6}} \right] = 0.19 \times \frac{0.45}{16.7 \times 10^{-6}} \times 10^{-3}$$

$$y^+ = 5.13 < y_{csL}^+$$

Wire is therefore within the conduction sub layer

$$q_0'' \text{ at wire} = q_0'' \text{ at wall}$$

$$q_0'' = -k \left(\frac{\partial T}{\partial y} \right)_{y=0} \quad \text{as } q_0'' = \text{const} \Rightarrow \frac{\partial T}{\partial y} = \text{const}$$

$$q_0'' = 684.21 \text{ W/m}^2$$

(b) Colburn's analogy: $St_x Pr^{2/3} = \frac{1}{2} C_{f,x}$

$$St_x = \frac{h}{\rho c_p u_\infty} = \frac{q_0''}{\rho c_p \Delta T u_\infty} \quad [\text{from part (i)}]$$

$$St_x = \frac{684.21}{1.13 \times 20 \times 10^3 \times 10} = 3 \times 10^{-3}$$

$$St_x = Pr^{-2/3} \cdot \frac{1}{2} C_{f,x} = (0.7)^{-2/3} \cdot \left(\frac{\tau_w}{\rho u_\infty^2} \right) [\text{Colburn}]$$
$$= 2.58 \times 10^{-3}$$

$$\% \text{ difference} = \frac{3 - 2.58}{3} = \frac{0.42}{3} \approx 16\%$$

Colburn analogy gives 16% error

(c) At centre of thermocouple, $y = 4.5 \text{ mm}$

$$y^+ = y \left(\frac{u^+}{\nu} \right) = 0.0045 \times \left(\frac{0.45}{16.7 \times 10^{-6}} \right) = 121.26 > y_{csL}^+$$

At top of spherical head, $y = 4.525 \text{ mm}$

$$y^+ = y \left(\frac{u^+}{v} \right) = 0.004525 \left(\frac{0.45}{16.7 \times 10^{-6}} \right)$$

At bottom of spherical head, $y = 4.475 \text{ mm}$

$$y^+ = y \left(\frac{u^+}{v} \right) = 0.004475 \left(\frac{0.45}{16.7 \times 10^{-6}} \right)$$

In the CTR, the temp profile is expressed as,

$$T^+ = Pr y_{csL}^+ + \frac{Pr_T}{K} \ln \frac{y^+}{y_{csL}^+}$$

$$T^+ = 14.10 \quad \text{where} \quad T^+ = (T_0 - \bar{T}) \frac{e C_p u_{\infty}}{q_0''}$$

$$\bar{T} = T_0 - \frac{T^+ q_0''}{e C_p u_{\infty}} = 11^\circ \text{C}$$

(d) $\frac{T_{\text{bottom}} - T_{\text{top}}}{T_{\text{mean}}} = ? \quad [\text{size-induced uncertainty}]$

$$\Delta T = \left(\frac{\partial T}{\partial y} \right) \Delta y \Rightarrow T_z - 11^\circ \text{C} = \frac{\partial T^+}{\partial y} \cdot \frac{q_0''}{e C_p u_{\infty}} (0.025)$$

$$T_b = 11^\circ \text{C} + \frac{q_0''}{e C_p u_{\infty}} \left[\frac{2.195}{y^+} \cdot \frac{\partial y^+}{\partial y} \right] (0.025) \approx 11.016^\circ \text{C}$$

$$T_b - 11^\circ \text{C} = \frac{\partial T^+}{\partial y} \cdot \frac{q_0''}{e C_p u_{\infty}} (-0.025)$$

$$T_b = 11^\circ \text{C} - \frac{q_0''}{e C_p u_{\infty}} \left[\frac{2.195}{y^+} \cdot \frac{\partial y^+}{\partial y} \right] (-0.025)$$

$$T_b \approx 10.984^\circ \text{C}$$

$$11 - 0.016$$

$$\frac{\Delta T}{\bar{T}} = \frac{0.032}{11}$$

(e) $u_{\infty} : 10 \text{ m/s} \rightarrow 50 \text{ m/s}$

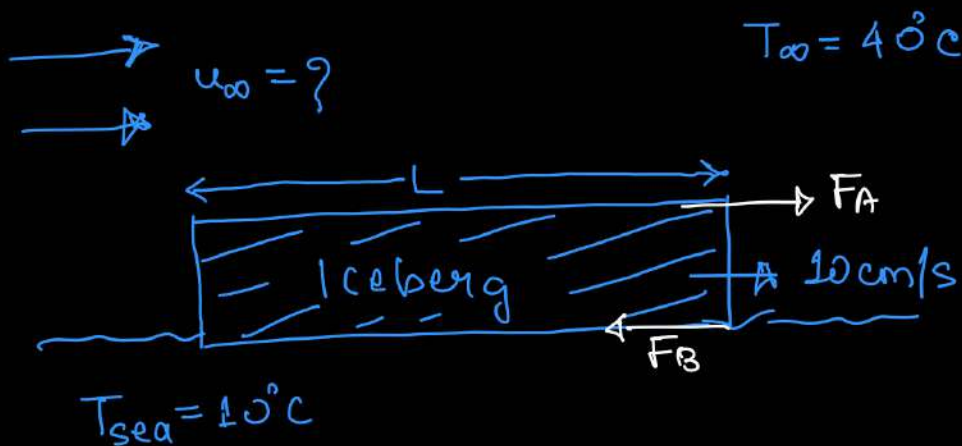
In first case, T_{measured} does not change

In second case, Pr_T changes and hence

T_{measured} would change

Problem 4

A flat sheet (tabular) iceberg drifts over the ocean as it is driven by the wind that blows over the top. The temperature of the surrounding seawater is 10°C , and the relative velocity between it and the iceberg is 10 cm/s . The length of the iceberg in the direction of drift is L . Calculate the corresponding wind velocity when the atmospheric air temperature is 40°C .



$F_A = F_B$ (as iceberg moves with const velocity)

$$\tau_{\text{air}} A = \tau_w A \quad \tau_{\text{avg}} = 0.37 \rho \cdot u^2 \left(\frac{uL}{\nu} \right)^{-1/5}$$

$$0.37 \rho_{\text{air}} u_{\text{air}}^2 \left(\frac{u_{\text{air}} L}{\nu_{\text{air}}} \right)^{-1/5} = 0.37 \rho_{\text{water}} u_{\text{water}}^2 \left(\frac{u_{\text{water}} L}{\nu_{\text{water}}} \right)^{-1/5}$$

Problem 5

Air flows with velocity 3.24 m/s over the top surface of the flat iceberg discussed in Problem 7.14. The air temperature outside the boundary layer is 40°C , and the ice surface temperature is 0°C . The length of the iceberg in the direction of air flow is $L = 100\text{ m}$. The ice latent heat of melting is $h_{sf} = 333.4\text{ kJ/kg}$.

Calculate the L -averaged heat flux deposited by the air flow into the upper surface of the iceberg (model this surface as flat). Calculate, in millimeters per hour, the rate of melting caused by this heat flux, that is, the erosion (thinning) of the ice slab.

$$q_0 = h(T - T_{\text{air}}) = ?$$

$$\downarrow$$

$$Nu = \frac{hL}{k} \Rightarrow h = \frac{Nu \cdot k}{L}$$

$$St_x Pr^{2/3} = \frac{1}{2} C_{f,x}$$

$$\boxed{\text{Ans: } 2.6 \text{ mm/hr}}$$

NATURAL CONVECTION

$$\boxed{\text{Volumetric Strain Rate: } \frac{DV}{Dt}}$$

$$\boxed{\text{Incompressible flow: } \frac{DV}{Dt} = 0}$$

$$\boxed{\frac{1}{V} \frac{DV}{Dt} = \vec{\nabla} \cdot \vec{v}}$$

$$m = \rho V \Rightarrow \ln m = \ln \rho + \ln V$$

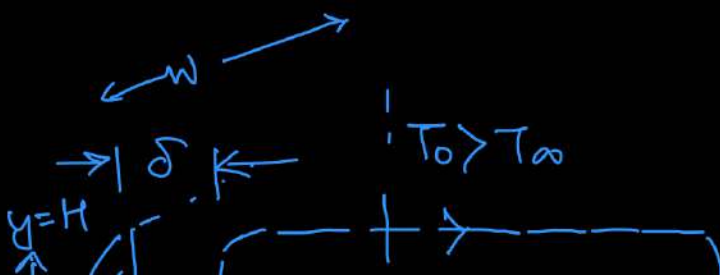
Taking substantial derivative,

$$\frac{1}{m} \frac{Dm}{Dt} = \frac{1}{\rho} \frac{D\rho}{Dt} + \frac{1}{V} \frac{DV}{Dt}$$

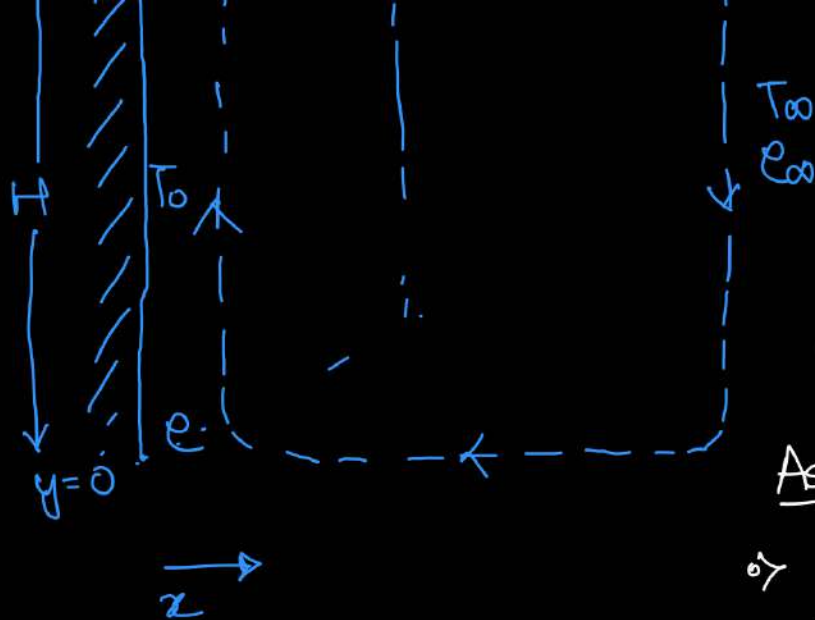
$\downarrow$ conservation of mass $\downarrow$ incompressible fluid

$$\frac{D\rho}{Dt} = 0 \quad [\text{incompressible fluid \& no mass transfer}]$$

$$\frac{\partial \rho}{\partial t} + u \frac{\partial \rho}{\partial x} + v \frac{\partial \rho}{\partial y} = 0$$



→ How much heat is transferred?



→ Can we control the velocity/motion wrt T_0 ?

$$q_0 = h_{avg} (W \times H) (T_0 - T_\infty)$$

Assumptions

- 2D flow
- Incompressible
- $H \gg \delta$

Continuity Eqn

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad \text{--- (i)}$$

Momentum Balance

$$\rho \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) = - \frac{\partial P}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$$

$$\rho \left(u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} \right) = - \frac{\partial P}{\partial y} + \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) - \rho g$$

Heat Balance

$$\rho c_p \left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} \right) = k \left[\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right]$$

For this situation,

$$\rho \left(u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} \right) = - \frac{\partial P}{\partial y} + \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) - \rho g$$

$$\rho c_p \left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} \right) = k \left[\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right] \quad \text{--- (ii)}$$

$$\frac{dP_\infty}{dy} = - \rho_\infty g$$

$$\rho \left(u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} \right) = (\rho_\infty - \rho) g + \mu \frac{\partial^2 v}{\partial x^2} \quad \text{--- (iii)}$$

Taking e as a function of T ,

$$e = e_m + \left. \frac{\partial e}{\partial T} \right|_{T=T_m} (T - T_m)$$

$$(e - e_\infty) = \left. \frac{\partial e}{\partial T} \right|_{T=T_\infty} (T - T_\infty)$$

$$\left. \frac{\partial e}{\partial T} \right|_{T=T_\infty} = \beta_\infty e_\infty$$

Volume Expansion

$$\begin{aligned} \text{Coeff } \beta &= \frac{1}{V} \frac{\partial V}{\partial T} \\ &= \frac{1}{1/e} \frac{\partial (1/e)}{\partial T} \\ &= -\frac{1}{e} \frac{\partial e}{\partial T} \end{aligned}$$

$$(e - e_\infty) = -\beta_\infty e_\infty (T - T_\infty)$$

Substituting this correlation in momentum balance

$$e \left(u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} \right) = \beta_\infty g (T - T_\infty) + \mu \frac{\partial^2 v}{\partial x^2}$$

Boussinesq Approximation: The density related to driving force is the only parameter which is a function of temperature.

$\mu, \alpha, e_{\text{lhs}}$ which would have been functions of temp are now constant.

$$e \left(u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} \right) = \beta_\infty g (T - T_\infty) + \mu \frac{\partial^2 v}{\partial x^2} \quad \text{--- ii)}$$

$$u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = \alpha \frac{\partial^2 T}{\partial x^2} \quad \text{--- iii)}$$

SCALING ANALYSIS

$$O(y) \sim H$$

$$O(x) \sim \delta_T$$

$$\rho(y) \sim \rho_0$$

$$o(u) \sim u$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \rightarrow \frac{u}{\delta_T} \sim \frac{v_c}{H} \rightarrow \boxed{u \sim v_c \left(\frac{\delta_T}{H} \right)}$$

$$u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = \alpha \frac{\partial^2 T}{\partial x^2} \quad [\text{Energy Balance}]$$

$$\Rightarrow v_c \frac{\delta_T}{H} \cdot \frac{\Delta T}{\delta_T} \quad v_c \frac{\Delta T}{H} \sim \alpha \frac{\Delta T}{\delta_T^2}$$

$$\Rightarrow \frac{v_c}{H} \sim \frac{\alpha}{\delta_T^2} \quad \boxed{v_c \sim \frac{\alpha H}{\delta_T^2}}$$

$$u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} = \nu \frac{\partial^2 v}{\partial x^2} + \beta \alpha g (T - T_\infty) \quad [\text{MOMENTUM BALANCE}]$$

$$\underbrace{v_c \frac{\delta_T}{H} \frac{v_c}{\delta_T} \quad v_c \frac{v_c}{H}}_{\frac{v_c^2}{H}} \quad \mu \frac{v_c}{\delta_T^2} \quad \beta \alpha g (\Delta T)$$

viscous force Buoyancy force

Scale everything in terms of buoyancy force

$$\frac{\text{friction}}{\text{buoyancy}} = \frac{\nu \frac{v_c}{\delta_T^2}}{g \beta \Delta T} = \frac{\nu \frac{\alpha H}{\delta_T^2 \alpha}}{g \beta \Delta T} = \left[\frac{\nu \alpha}{g \beta \Delta T H^3} \right] \cdot \left(\frac{H^4}{\delta_T^4} \right)$$

$$1 = \frac{1}{Ra_H} \cdot \left(\frac{H}{\delta_T} \right)^4 \quad [Ra_H: \text{Rayleigh No}]$$

$$\frac{\text{Inertia}}{\text{Buoyancy}} = \frac{v_c^2/H}{g \beta \Delta T} = \frac{\alpha^2 H^2}{\delta_T^4} \cdot \frac{1}{H}$$

$$= \left(\frac{\alpha \nu}{g \beta \Delta T H^3} \right) \cdot \left(\frac{\alpha}{\nu} \right) \cdot \left[\frac{H}{\delta_T} \right]^4$$

$$1 = \frac{1}{Pr} \cdot \frac{1}{Ra_H} \cdot \left[\frac{H}{\delta_T} \right]^4$$

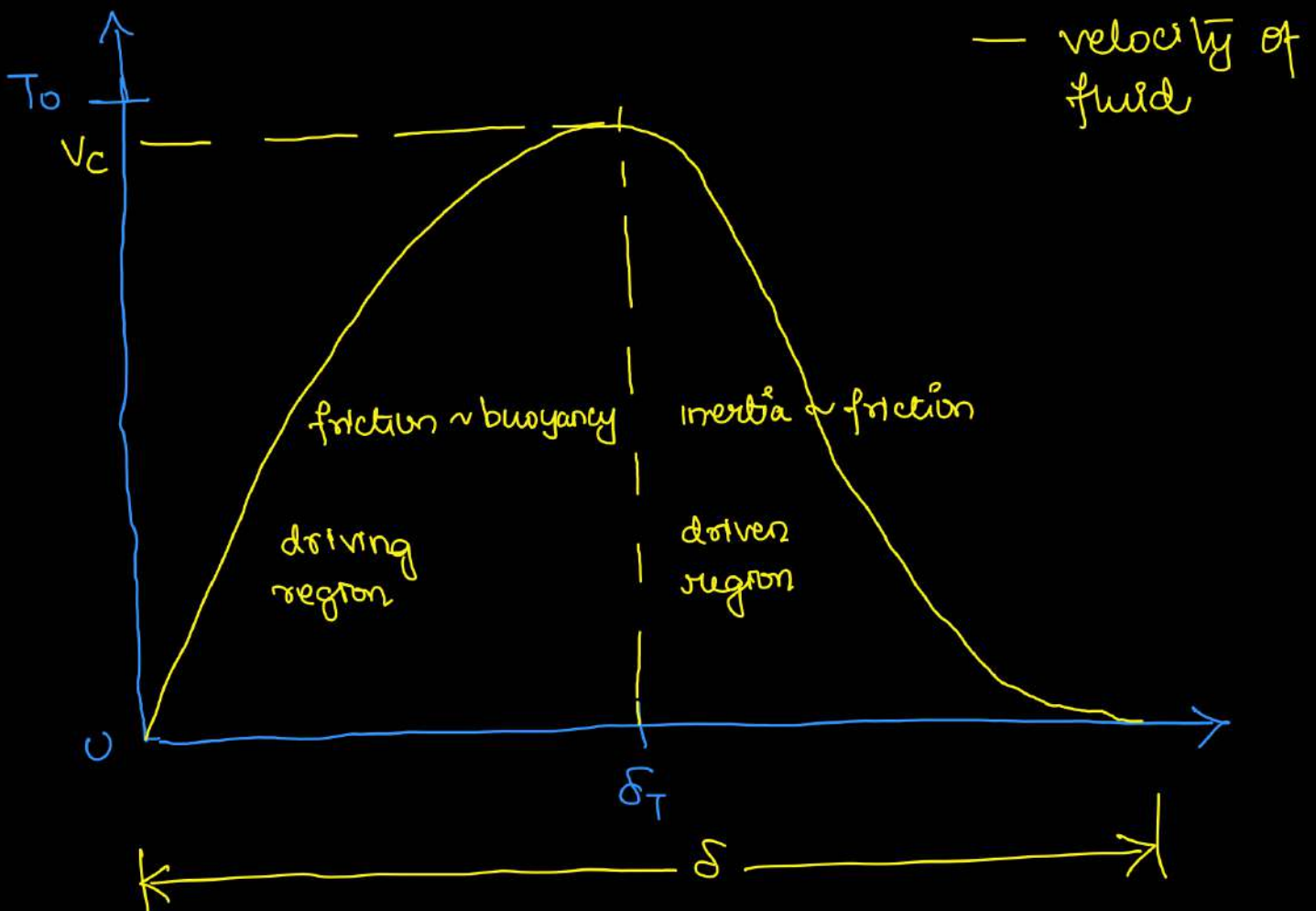
(A) HPgh Pr fluid

$$Ra_H^{-1} \left(\frac{H}{\delta_T} \right)^4 \sim 1 \Rightarrow \boxed{\delta_T \sim H Ra_H^{-1/4}}$$

$$h \sim \frac{K}{\delta_T} \sim \frac{K}{H} Ra_H^{1/4}$$

$$\boxed{Nu \sim Ra_H^{1/4}}$$

$$V_c \sim \frac{\alpha H}{\delta_T^2} \sim \frac{\alpha H}{H Ra_H^{-1/4}} \sim \frac{\alpha Ra_H^{1/2}}{H}$$



$$\frac{V_c^2}{H} \sim \frac{v V_c}{\delta^2} \Rightarrow \frac{V_c}{H} \sim \frac{v}{\delta^2} \Rightarrow \delta^2 \sim \frac{v H}{V_c}$$

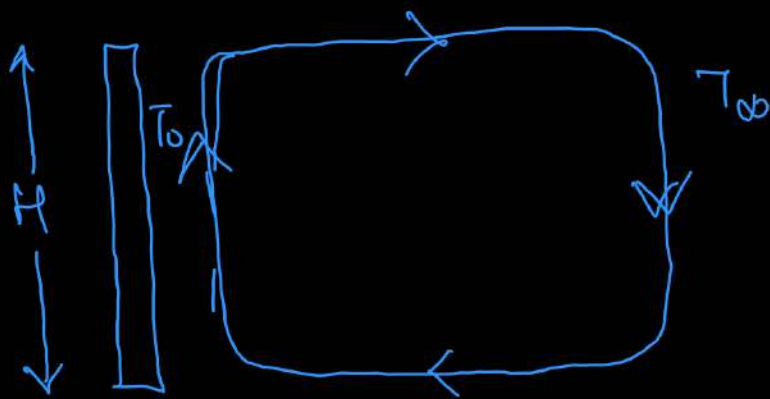
$$\Rightarrow \delta^2 \sim \frac{v H}{\frac{\alpha}{H} Ra_H^{1/2}} \Rightarrow \boxed{\delta \sim H Pr^{1/2} Ra_H^{-1/4}}$$

Comparing δ and δ_T ,

$$\boxed{\frac{\delta_T}{\delta} \sim Pr^{-1/2}}$$

Pr high $\Rightarrow \delta_T \ll \delta$.

19/03/24

FORCED VS NATURAL CONVECTION v_c characteristic velocity y -comp momentum balance,

$$u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} = \nu \frac{\partial^2 v}{\partial x^2} + g \beta_m (T - T_m)$$

Non-dimensionalizing the eqn,

$$u^* = \frac{u}{v_c} \quad v^* = \frac{v}{v_c} \quad x^* = \frac{x}{H} \quad \Theta = \frac{(T - T_\infty)}{(T_0 - T_\infty)}$$

$$\frac{v_c^2}{H} u^* \frac{\partial v^*}{\partial x^*} + \frac{v_c^2}{H} v^* \frac{\partial v^*}{\partial y^*} = \frac{v_c}{H^2} \nu \frac{\partial^2 v^*}{\partial x^{*2}} + g \beta_m (T_0 - T_m) \Theta$$

$$u^* \frac{\partial v^*}{\partial x^*} + v^* \frac{\partial v^*}{\partial y^*} = \underbrace{\frac{1}{v_c H} \nu \frac{\partial^2 v^*}{\partial x^{*2}}}_{\text{viscous term controllable}} + \underbrace{g \beta_m \frac{H}{v_c^2} (T_0 - T_m) \Theta}_{\text{Buoyancy term not controllable}}$$

$$\left(\frac{g \beta_m (T_0 - T_m) H^3}{\nu^2} \right) \cdot \left(\frac{H \nu^2}{v_c^2 H^3} \right) = \frac{Gr}{Re^2} = Ri$$

Gr: Grashoff (?) No.

Ri: Richardson (?) No

 $Re \ll 1$ Natural Convection

$$Pr \gg 1$$

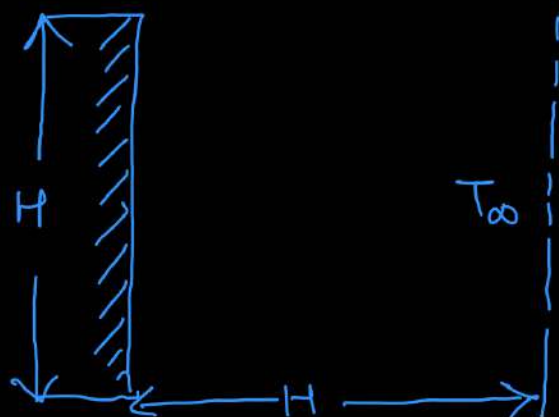
$$Pr \ll 1$$

Forced Convection

[Integral Soln provided in book.]

Problem 1

Consider a vertical wall of height H in contact with an isothermal fluid reservoir. For the purpose of scale analysis, select the square flow region of height H and horizontal thickness H . Show that if in the momentum equation you invoke a balance between friction and buoyancy, the inertia/friction ratio comes out to be of order $Gr_H = (g\beta \Delta T H^3)/\nu^2$ (note that the $H \times H$ region is not the boundary layer region; hence, the conclusion "inertia/friction $\sim Gr_H$ " does not apply to the boundary layer region). Is the vertical velocity scale derived above compatible with the v scale recommended by the energy equation for the $H \times H$ region? In other words, is the invoked balance friction $\sim$ buoyancy in the $H \times H$ region realistic?



$$\begin{aligned} x &\sim H & u &\sim v_c \\ y &\sim H & v &\sim v_c \\ \text{Momentum,} & & & \\ \underbrace{u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y}}_{\frac{v_c^2}{H}} &= & \underbrace{\nu \frac{\partial^2 v}{\partial x^2}}_{\frac{\nu v_c}{H^2}} + g\beta m(T - T_\infty) & \sim g\beta \Delta T \end{aligned}$$

$$\frac{\nu v_c}{H^2} \sim g\beta \Delta T$$

$$\Rightarrow \boxed{v_c \sim \frac{g\beta \Delta T H^2}{\nu}}$$

$$Gr = \frac{\text{Inertia}}{\text{Friction}} = \frac{v_c^2/H}{\nu v_c/H^2} = \frac{H v_c}{\nu} = \frac{H^3 g\beta \Delta T}{\nu^2}$$

Energy,

$$u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = \alpha \frac{\partial^2 T}{\partial x^2}$$

$$v_c \frac{\Delta T}{H}$$

$$\propto \frac{\Delta T}{H^2}$$

$$v_c \frac{\Delta T}{H} \sim \propto \frac{\Delta T}{H^2}$$

$$v_c \sim \frac{\alpha}{H}$$

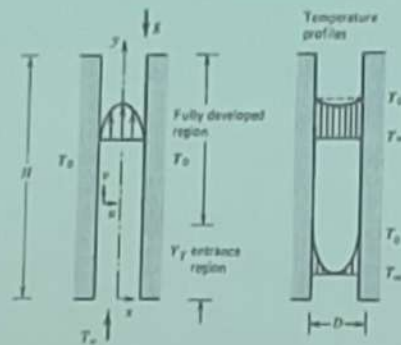
→ diff v_c scales
for energy &
momentum

Horizontal scale H is not a correct assumption
which leads to two different scales for v_c
based on 2 eqns - momentum & balance.

26/03/24

Using natural flows, a solar chimney has to be designed to heat the air to a uniform temperature at a constant mass flow rate. The chimney is planned to design using two parallel walls with the width of the wall significantly higher than the height of the chimney. Design the chimney by looking into the H/D . What is the condition for which the designed chimney will work and will be able to discharge fluid at a constant rate? Assume any necessary conditions with proper justification.

Hint: Find the average heat flux between the two parallel walls facing each other as depicted in the figure. Assume the necessary conditions for an analytical solution to the problem. Validate your assumptions with the necessary condition



Problem 2: (Previous Class)

$$m = \frac{g \beta e (T_0 - T_\infty) D^3 W}{12 \nu}$$

$$q = m c_p (T_0 - T_\infty)$$

$$q''_{\text{avg}} = \frac{q}{2HW} \rightarrow \text{area}$$

$$= \frac{g \beta e (T_0 - T_\infty)^2 D^3 c_p}{24 H \nu}$$

$$Nu = h_{\text{avg}} H = \frac{q_{\text{tot}}}{H} = \frac{R_{\text{ad}}}{H}$$

1ST ASSUMPTION

$$\gamma_T \ll H \quad (\text{assumption})$$

$$\underline{Pr \gg 1}$$

$$\delta_T \sim H Ra_H^{-1/4}$$

$$\frac{D}{2} \sim \gamma_T Ra_{\gamma_T}^{-1/4}$$

$$\left(\frac{\gamma_T}{D}\right)^4 \sim \left(\frac{1}{2}\right)^4 Ra_{\gamma_T}$$

$$\left(\frac{D}{\gamma_T}\right)^3 = \frac{Ra_D}{Ra_{\gamma_T}} \Rightarrow Ra_{\gamma_T} = Ra_D \left(\frac{\gamma_T}{D}\right)^3$$

Substituting γ_T as it is unknown,

$$\left(\frac{\gamma_T}{D}\right)^4 \sim \left(\frac{1}{2}\right)^4 Ra_D \left(\frac{\gamma_T}{D}\right)^3$$

$$\gamma_T \sim \left(\frac{1}{2}\right)^4 D Ra_D$$

$$D Ra_D \left(\frac{1}{2}\right)^4 \ll H \quad (\gamma_T \ll H)$$

$$Ra_D \ll 2^4 \left(\frac{H}{D}\right)$$

$$\boxed{Ra_D^{1/4} \ll 2 \left(\frac{H}{D}\right)^{1/4}}$$

} Geometry must hold

2ND ASSUMPTION

$$(T_0 - T) \ll (T_0 - T_\infty)$$

$$q''_{\text{over}} \sim \frac{k(T_0 - T)}{D} \Rightarrow (T_0 - T) \sim q''_{\text{over}} \frac{D}{k}$$

$$q''_{\text{over}} \frac{D}{k} \ll (T_0 - T_\infty)$$

$$q''_{\text{over}} \sim Ra_D \Delta T \frac{k}{H}$$

$$\left[Nu = \frac{q''_{\text{over}} H}{\Delta T k} = \frac{Ra_D}{24} \right]$$

Substituting q_{over} ,

$$Ra_D \propto \frac{k}{H} \cdot \frac{D}{k} \ll (T_0 - T_\infty)$$

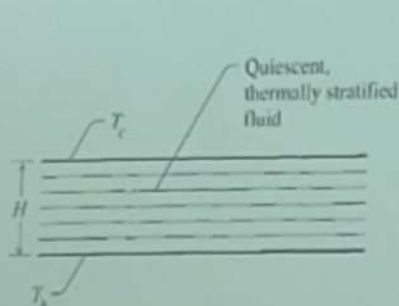
$$\boxed{Ra_D^{1/4} \ll \left(\frac{H}{D}\right)^{1/4}}$$

} same assumption.

INTERNAL NATURAL CONVECTION

Internal Natural Convection

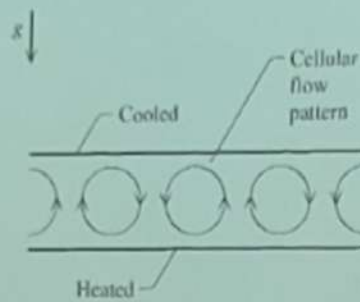
- (1) enclosures heated from the side.
- (2) enclosures heated from below.



$$Ra_H < 1708$$

$$\bar{Nu}_H = 1$$

(a)



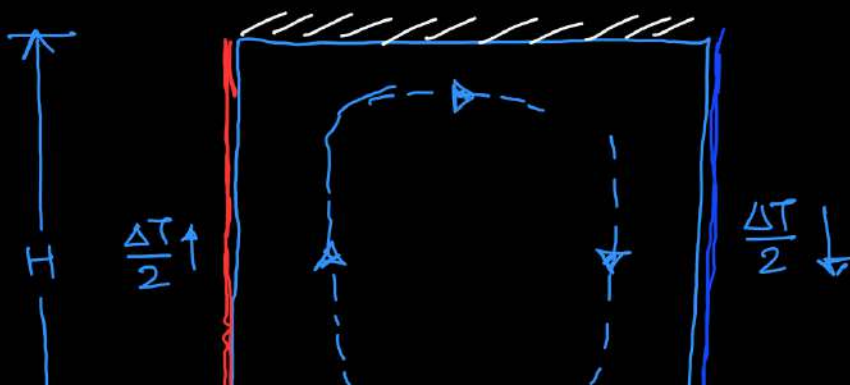
$$Ra_H > 1708$$

$$\bar{Nu}_H > 1$$

(b)

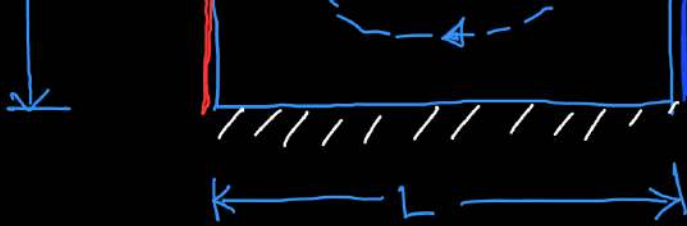
Critical Rayleigh number

$$Ra_H \gtrsim 1708$$



Assumptions

- Boussinesq Assumption
- Incompressible



Continuity Eqn : $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$ — (i)

x: Momentum Eqn : $\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{\partial P}{\partial x} + \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$ — (ii)

y: Momentum Eqn : $\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} = -\frac{1}{\rho} \frac{\partial P}{\partial y} + \nu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) + g\beta[1 - \beta(T - T_0)]$ — (iii)

Energy Eqn : $\frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = \alpha \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right)$ — (iv)

$g\beta[1 - \beta(T - T_0)] \rightarrow \rho = \rho_0 - \beta\rho_0(T - T_0)$
Expand using Taylor's series

$\frac{\partial}{\partial y} \left(\text{x-momentum balance eqn} \right) : \quad \text{--- (v)}$

$$\frac{\partial}{\partial y} \cdot \frac{\partial u}{\partial t} + u \frac{\partial^2 u}{\partial x \partial y} + v \frac{\partial^2 u}{\partial y^2} = -\frac{1}{\rho} \frac{\partial^2 P}{\partial x \partial y} + \frac{\partial}{\partial y} \left[\nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) \right]$$

$\frac{\partial}{\partial x} \left[\text{y-momentum balance eqn} \right] \quad \text{--- (vi)}$

$$\frac{\partial}{\partial x} \cdot \frac{\partial v}{\partial t} + u \frac{\partial^2 v}{\partial x^2} + v \frac{\partial^2 v}{\partial x \partial y} = -\frac{1}{\rho} \frac{\partial^2 P}{\partial x \partial y} + \frac{\partial}{\partial x} \left[\nu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) \right] + g\beta \frac{\partial T}{\partial x}$$

(vi) - (v),

$$\left[\frac{\partial}{\partial x} \frac{\partial v}{\partial t} + u \frac{\partial^2 v}{\partial x^2} + v \frac{\partial^2 v}{\partial x \partial y} \right] - \left[\frac{\partial}{\partial y} \frac{\partial u}{\partial t} + u \frac{\partial^2 u}{\partial x \partial y} + v \frac{\partial^2 u}{\partial y^2} \right] = -\frac{1}{\rho} \frac{\partial^2 P}{\partial x \partial y} + \frac{\partial}{\partial x} \left[\nu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) \right] - \frac{\partial}{\partial y} \left[\nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) \right] + g\beta \frac{\partial T}{\partial x}$$

$$= \frac{\partial}{\partial x} \left[\underset{\nu}{\nu} \left(\underset{\nu}{\frac{\partial^2 v}{\partial x^2}} + \underset{\nu}{\frac{\partial^2 v}{\partial y^2}} \right) \right] - \frac{\partial}{\partial y} \left[\underset{\nu}{\nu} \left(\underset{\nu}{\frac{\partial^2 u}{\partial x^2}} + \underset{\nu}{\frac{\partial^2 u}{\partial y^2}} \right) \right] + \underset{b}{g\beta \frac{\partial T}{\partial x}}$$

ρ : inertial

ν : viscous

b : buoyancy

Scaling Analysis

$$y \sim H$$

$$u \sim u$$

$$t \sim t$$

$$v \sim v_0$$

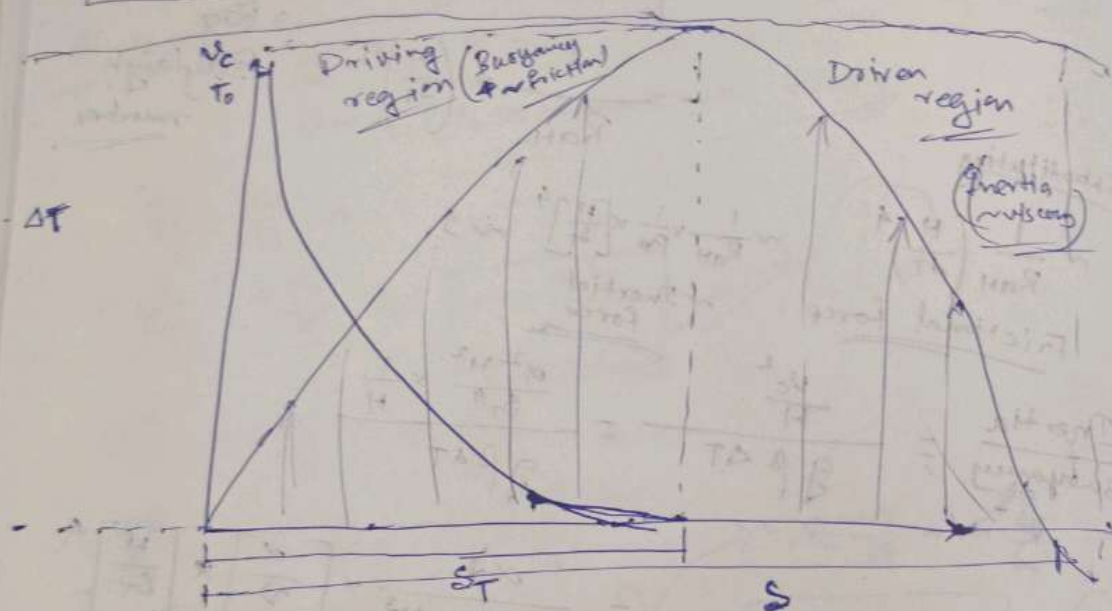
$$x \sim \delta_T \quad (t \sim 0)$$

$$x \sim L \quad (t \gg 0)$$

Energy Equation

$$v_c \sim \frac{\alpha H}{S_T}$$

$$v_c \sim \frac{\alpha}{H} Ra_H^{1/2}$$



Inertia $\rightarrow$ Driving the fluid $x \rightarrow$ (far away from wall)
viscous $\rightarrow$ Resist the flow of fluid.

Driven region $\left| \frac{v_c^2}{H} \sim \nu \frac{v_c}{S_T^2} \right|$

$$\frac{v_c}{H} \sim \frac{\nu}{S_T^2}$$

$$S_T^2 = \frac{H\nu}{v_c}$$

$$S_T = \frac{H\nu}{\alpha/H Ra_H^{1/2}} \Rightarrow H^2 Pr Ra_H^{-1/2}$$

$$S_T \sim H Pr^{1/2} Ra_H^{-1/4}$$

$$\frac{S_T}{S} = \frac{H Ra_H^{-1/2}}{H Pr^{1/2} Ra_H^{-1/4}}$$

$$u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} =$$

$$u^* = \frac{u}{v_c} \quad ; \quad v^* = \frac{v}{v_c}$$

$$u = u^* v_c$$

$$\frac{v_c^2}{H} u^* \frac{\partial v^*}{\partial x^*}$$

$$u^* \frac{\partial v^*}{\partial x^*} + v^* \frac{\partial v^*}{\partial y^*} =$$

$$g \beta_\Delta (T_0 - T_1)$$

$$g \beta_\Delta \frac{(T_0 - T_1)}{H}$$

When the

$$= 1 \rightarrow$$

$$\frac{\delta_T}{\delta} = \frac{H Ra_h^{-1/4}}{H Pr^{1/2} Ra_h^{-1/4}} \Rightarrow \underline{Pr^{-1/2}}$$

$$\boxed{\delta_T < \delta}$$

Forced vs Natural Convection

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = \nu \frac{\partial^2 u}{\partial x^2} + g \beta_\infty (T - T_\infty) \quad \text{13/3/24}$$

vertical (y-coord) momentum balance eqn.

$$u^* = \frac{u}{v_c} \quad ; \quad v^* = \frac{v}{v_c} \quad ; \quad x^* = \frac{x}{H}$$

$$u = u^* v_c \quad ; \quad v = v^* v_c \quad ; \quad y^* = \frac{y}{H}$$

$$\theta = \frac{T - T_\infty}{T_o - T_\infty}$$

$$\frac{v_c^2}{H} u^* \frac{\partial v^*}{\partial x^*} + \frac{v_c^2}{H} v^* \frac{\partial v^*}{\partial y^*} = \frac{v_c}{H^2} \nu \frac{\partial^2 v^*}{\partial x^{*2}} + g \beta_\infty (T_o - T_\infty) \theta$$

$$u^* \frac{\partial v^*}{\partial x^*} + v^* \frac{\partial v^*}{\partial y^*} = \frac{H}{v_c^2} \times \frac{v_c}{H^2} \nu \frac{\partial^2 v^*}{\partial x^{*2}} + \frac{H}{v_c^2} g \beta_\infty (T_o - T_\infty) \theta$$

If this coefficient is significantly high then the buoyancy force is significantly high.

for a given system this can't be controlled.

$$g \beta_\infty \frac{(T_o - T_\infty) H^3}{\nu^2} \times \frac{H^2 \nu^2}{v_c^2 H^3} \quad \left(\begin{array}{l} \text{Divide by } \nu^2 \text{ multiply by } \nu^2 \\ \text{To make Grashoff number.} \end{array} \right)$$

$$g \beta_\infty \frac{(T_o - T_\infty) H^3}{\nu^2} \times \frac{\nu^2}{v_c^2 H^2}$$

$$\sim \frac{Gr_o}{Re^2} \sim Ri = \text{Richardson Number}$$

When this Richardson Number is $\gg 1$, $\rightarrow$ lead to free convection
 $Re < 1 \rightarrow$ buoyancy effect neglected forced convection.
 mixed convection.

Integral Solution

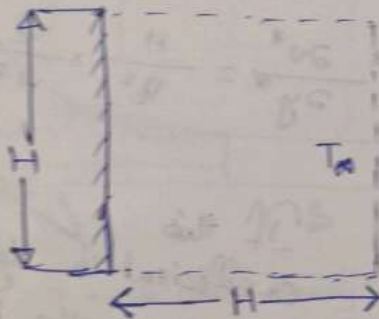
$$\frac{d}{dy} \int_0^x u^2 dx = -v \left(\frac{\partial u}{\partial x} \right)_{x=0} + g\beta \int_0^x (T - T_\infty) dx \quad (1)$$

$$\frac{d}{dy} \int_0^x v (T_\infty - T) dx$$

Assume a velocity profile & Temperature profile.
 High Pr fluids & low Pr fluids (Note)

Similarity Solutions (From book)

Problems 1



Thermals = convection

eqn

$$u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} = v \frac{\partial^2 v}{\partial x^2} + g\beta (\Delta T)$$

Scale of x for this system

$$x \sim H; \quad y \sim H; \quad v \sim v_c; \quad u \sim u_c.$$

$$u_c \frac{\partial v_c}{\partial H} + v_c \frac{\partial v_c}{\partial H} =$$

$$\frac{v_c^2}{H}$$

$$\frac{v_c^2}{H^2}$$

$$g\beta (\Delta T)$$

$$\frac{v_c^2}{H^2} \sim g\beta \Delta T$$

$$\Rightarrow v_c \sim \frac{g\beta \Delta T H^2}{\nu}$$

Chapter 5

Integral Solutions

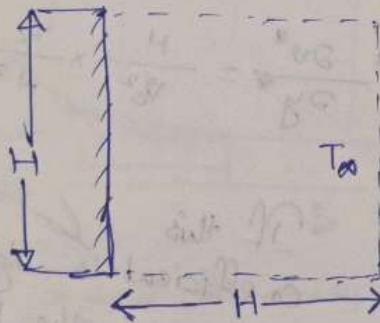
$$\frac{d}{dy} \int_0^x v^2 dx = -v \left(\frac{\partial v}{\partial x} \right)_{x=0} + g\beta \int_0^x (T - T_\infty) dx \quad (1)$$

$$\frac{d}{dy} \int_0^x v (T_\infty - T) dx$$

Assume a velocity profile & Temperature profile.
High Pr fluids & low prandtl fluids (Note)

Similarity Solutions (from book)

Problems 1



eqn

$$u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} = \nu \frac{\partial^2 v}{\partial x^2} + g\beta (\Delta T)$$

scale of x for this system

$$x \sim H; \quad y \sim H; \quad v \sim v_c; \quad u \sim u_c$$

$$u_c \frac{\partial v_c}{\partial H} + v_c \frac{\partial v_c}{\partial H} =$$

$$\frac{v_c^2}{H}$$

$$\frac{v_c^2}{H^2}$$

$$g\beta(\Delta T)$$

$$\frac{v_c^2}{H^2} \sim g\beta\Delta T$$

$$\Rightarrow v_c \sim \frac{g\beta\Delta TH^2}{\nu}$$

equation
ine
fric

If we

If we

Part 2

con

2 eqn an

Chapter 7

Chapter 4

Chapter 5

equating inertia & friction to get Grashoff no.

$$\frac{\text{inertia}}{\text{friction}} \sim \frac{v_c^2/H}{\frac{\mu v_c}{H^2}} \sim \frac{v_c H}{\mu}$$

If we multiply this,

If we substitute v_c with its value we get Grashoff no.

Part 2

conduction ~ convection. (from energy equation)

$$\sim \alpha \frac{\Delta T}{H^2} \sim \frac{v_c \Delta T}{H} \Rightarrow v_c \sim \frac{\alpha}{H}$$

2 eqn and only unknown is v_c , so we are getting two v_c .

Chapter 7

TBL

7.1, 7.2, 7.22 eqn, 7.4, 7.3, 7.41

7.69

Colburn analogy

7.81, 7.82

Solve

upto

7.8

(Initial 15 problem) Reynolds Analogy

Chapter 4

ENC

4.1

Bousinesq approximation

4.3

Scale Analysis, (4.3.1) High Pr

Table 4.1

4.3.1 fluids

4.4 (Integral solution + similarity soln)

4.6 (Uniform Heat flux)

upto page 192

4.9 as a problem & solved

4.10

described in different ways. (4.112)

4.11.2, 4.11.3,

Horizontal cylinder (Page 209)

4.1, 5.1, 5.8

What is the regime (5.8)

5.2

Integral soln X

2.49 (upto)

Shallow

Chapter 5

INC

$$\delta_T \sim \left(\frac{\alpha L}{u_{\infty}} \right)^{1/2} = \left(\frac{\alpha \nu L^2}{u_{\infty} \nu L} \right)^{1/2} = L \left(\frac{\alpha}{\nu} \right)^{1/2} \left(\frac{\nu}{u_{\infty} L} \right)^{1/2}$$

$$\frac{\delta_T}{L} \sim Pr^{-1/2} Re^{-1/2}$$

$$\delta_T \ll \delta$$

AHT (MC) [Natural Convection] 12/3/24

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$\rho \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) = \mu \frac{\partial^2 u}{\partial x^2} + g(\rho_{\infty} - \rho) \quad \text{--- ①}$$

$$\left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} \right) = \alpha \frac{\partial^2 T}{\partial x^2}$$

~~Assumption~~ $\rho = \rho_{\infty} + \frac{\partial \rho}{\partial T} \bigg|_{T=T_{\infty}} (T - T_{\infty}) + \text{higher order terms.}$

$$(\rho - \rho_{\infty}) = \frac{\partial \rho}{\partial T} \bigg|_{T=T_{\infty}} (T - T_{\infty}) + \dots$$

$$(\rho - \rho_{\infty}) = -\beta_{\infty} \rho_{\infty} (T - T_{\infty})$$

$$(\rho_{\infty} - \rho) = \beta_{\infty} \rho_{\infty} (T - T_{\infty})$$

Boussinesq Approximation,

Density is constant everywhere, except the location of driving force.

In the eqn ①,

$$\rho \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) = \mu \frac{\partial^2 u}{\partial x^2} + g(\rho_{\infty} - \rho)$$

expresses as ρ_{∞}

This is constant

$$\rho \left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} \right) = \alpha \frac{\partial^2 T}{\partial x^2}$$

Volume Expansion Co-efficient

$$\beta = \frac{1}{\rho} \frac{\partial \rho}{\partial T} \bigg|_T$$

$$= \frac{1}{(\rho_p)} \frac{\partial (\rho_p)}{\partial T}$$

$$= \frac{1}{\rho} \frac{\partial \rho}{\partial T}$$

$$\beta_{\infty} = -\frac{1}{\rho_{\infty}} \frac{\partial \rho}{\partial T} \bigg|_{T=T_{\infty}}$$

$$-\beta_{\infty} \rho_{\infty} = \frac{\partial \rho}{\partial T} \bigg|_{T=T_{\infty}}$$

This will be the ρ_{∞} of Temp. ρ/c to Boussinesq Approx.

Ans

$$\left(u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y}\right) = \nu \frac{\partial^2 v}{\partial x^2} + [g \beta_0 (T - T_a)]$$

$$\left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y}\right) = \alpha \frac{\partial^2 T}{\partial x^2}$$

Using Scaling Analysis we can solve this

$$Q = A_{max} \times h_{max}$$

$$h \sim \frac{k}{S_T} \text{ (scaling)}$$

$$y \sim H, x \sim S_T, u \sim u$$

$$v \sim v_c \text{ (} v_c \text{ is characteristic velocity)}$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$\sim \frac{u}{S_T} \sim \frac{v_c}{H} \Rightarrow \boxed{u \sim \frac{v_c S_T}{H}}$$

$$u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = \alpha \frac{\partial^2 T}{\partial x^2}$$

$$\sim u \frac{\Delta T}{S_T} \sim v_c \frac{\Delta T}{H}$$

$$\sim \alpha \frac{\Delta T}{S_T^2}$$

$$v_c \frac{\Delta T}{H} \sim \alpha \frac{\Delta T}{S_T^2}$$

$$\sim v_c \frac{\Delta T}{H}$$

$$\sim \alpha \frac{\Delta T}{S_T^2}$$

$$\boxed{v_c = \frac{\alpha H}{S_T^2}}$$

$$\left(u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y}\right) = \nu \frac{\partial^2 v}{\partial x^2} + g \beta (T - T_a)$$

$$\sim u \frac{v_c}{S_T}$$

$$\sim v_c \frac{v_c}{H}$$

$$\sim \nu \frac{v_c}{S_T^2}$$

$$\sim g \beta (\Delta T)$$

$$\sim \frac{v_c^2}{H}$$

$$\sim \nu \frac{v_c}{S_T^2}$$

$$\sim g \beta (\Delta T)$$

Inertial force

Frictional/viscous

Buoyancy

The Assumption for

natural convection is

buoyancy force has to be there

$$\frac{\text{Friction}}{\text{Buoyancy}} = \frac{\nu \frac{v_c}{S_T^2}}{g \beta \Delta T}$$

Substituting

$$\sim \frac{1}{RaH} \left(\frac{H}{\Delta T}\right)^4$$

Frictional force

$$\frac{\text{Inertia}}{\text{Buoyancy}} = \frac{\frac{v_c}{H}}{g \beta \Delta T}$$

For high Prandtl

$$RaH^{-1} \left(\frac{H}{S_T}\right)^4$$

$$\boxed{S_T = H}$$

Scale of H, $h \sim$

$$\boxed{Nu \sim R}$$

$$\frac{\text{Friction}}{\text{Buoyancy}} = \frac{\frac{\nu_c^2}{S_T^2}}{g \beta \Delta T} = \frac{\frac{\alpha H}{S_T^4}}{g \beta \Delta T}$$

$$= \left[\frac{\alpha}{g \beta \Delta T H^3} \right] \times \left(\frac{H^4}{S_T^4} \right)$$

$$= \frac{1}{Ra_H} \times \left(\frac{H}{S_T} \right)^4 \rightarrow \text{Rayleigh number}$$

substituting

$$\sim \frac{1}{Ra_H} \left(\frac{H}{\Delta T} \right)^4 \sim \frac{1}{Ra_H} \times \frac{1}{Pr} \times \left[\frac{H}{S_T} \right]^4 \sim 1.$$

Frictional force

Inertial force

$$\frac{\text{Inertia}}{\text{Buoyancy}} = \frac{\frac{\nu_c^2}{H}}{g \beta \Delta T} = \frac{\frac{\alpha^2 H^2}{S_T^4} \times \frac{1}{H}}{g \beta \Delta T}$$

$$= \frac{\alpha}{g \beta \Delta T H^3} \times \left[\frac{\alpha}{\nu} \right] \times \left[\frac{H}{S_T} \right]^4$$

$$= \frac{1}{Ra_H} \times \frac{1}{Pr} \times \left[\frac{H}{S_T} \right]^4$$

Hi for high Prandtl number fluid

$$Ra_H^{-1} \left(\frac{H}{S_T} \right)^4 \sim 1.$$

$$S_T = H Ra_H^{-1/4}$$

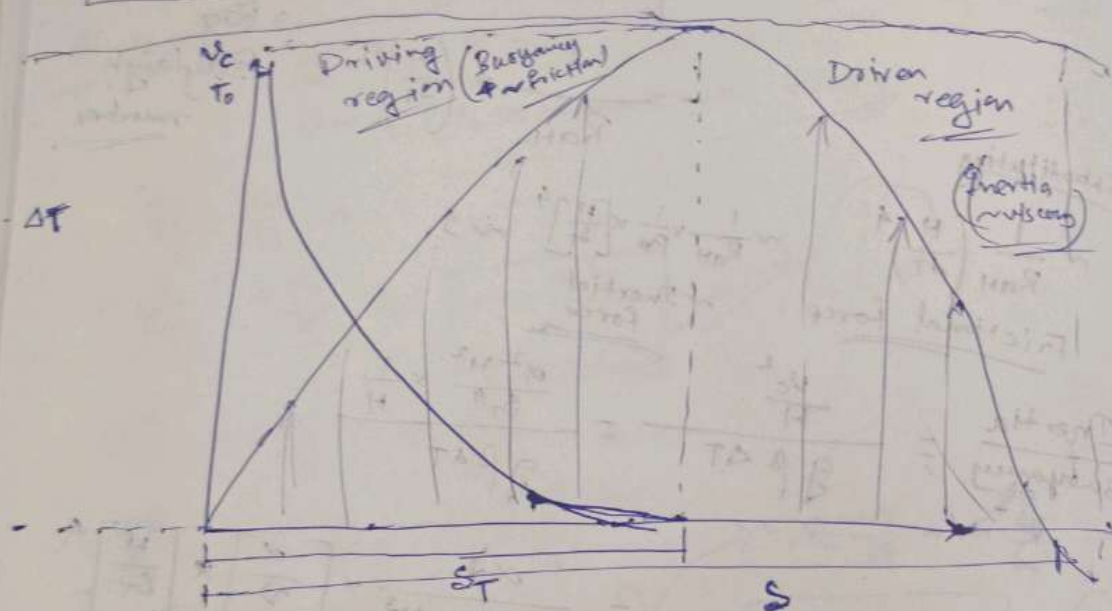
$$\text{Scale of } H, h \sim \frac{\kappa}{S_T} \sim \frac{\kappa}{H} Ra_H^{1/4}$$

$$Nu \sim Ra_H^{1/4}$$

Energy Equation

$$v_c \sim \frac{\alpha H}{S_T}$$

$$v_c \sim \frac{\alpha}{H} Ra_H^{1/2}$$



Inertia $\rightarrow$ Driving the fluid
viscous $\rightarrow$ Resist the flow of fluid

Driven region $\left| \frac{v_c^2}{H} \sim \nu \frac{v_c}{S_T^2} \right|$

$$\frac{v_c}{H} \sim \frac{\nu}{S_T^2}$$

$$S_T^2 = \frac{H\nu}{v_c}$$

$$S_T = \frac{H\nu}{\alpha/H Ra_H^{1/2}} \Rightarrow H^2 Pr Ra_H^{-1/2}$$

$$S_T \sim H Pr^{1/2} Ra_H^{-1/4}$$

$$\frac{S_T}{S} = \frac{H Ra_H^{-1/2}}{H Pr^{1/2} Ra_H^{1/2}}$$

$$u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} =$$

$$u^* = \frac{u}{v_c} \quad ; \quad v =$$

$$u = u^* v_c \quad ;$$

$$\frac{v_c^2}{H} u^* \frac{\partial v^*}{\partial x^*}$$

$$u^* \frac{\partial v^*}{\partial x^*} + v$$

$$g \beta_\Delta (T_0$$

$$g \beta_\Delta (T_0$$

When the

$$= 1 \rightarrow$$

Problem 1

You have a bottle of cold drink at room temperature, and you would like to drink it cold as soon as possible. You place the bottle in the refrigerator; however, you have the option of positioning the bottle (1) vertically or (2) horizontally. The refrigerator cools by natural convection (it does not employ forced circulation).

(a) Which way should you position the bottle? Describe the goodness of your decision by calculating the ratio t_1/t_2 , where t represents the order of magnitude of the time needed for the bottle to reach thermal equilibrium with the refrigeration chamber (base this calculation on scale analysis).

Data :

$$\left(\frac{g\beta}{\alpha\nu}\right)_{air} = 125; \quad \left(\frac{g\beta}{\alpha\nu}\right)_{cd} = 4910 \quad ; \text{ both have same unit}$$

$k_{air}=0.025$ and $k_{cd}=0.58$; both have same unit Subscript cd stands for *cold drink*

For high Pr number fluids (e.g. air, water *etc.*) the Nu scales as

For heat transfer from vertical walls

$$Nu = \frac{Hh}{k} \propto Ra_H^{1/4} \quad \text{Where, } Ra_H \text{ is the Rayleigh number } Ra_H = \frac{g\beta\Delta TH^3}{\alpha\nu} \quad \text{Where } H \text{ is the height of the wall}$$

For heat transfer from horizontal walls

$$Nu = \frac{Lh}{k} \propto Ra_L^{1/4} \quad \text{Where } L \text{ is the characteristic length}$$

Consider $H=5L$ and assume $[(T_{cd} - T_s) = (T_s - T_{air})]$ Where T_s is the surface temperature of the bottle

Solution:

Assumptions: (1) The bottle is cooled only by natural convection (2) No spatial variation of temperature, the temperature of the surface of the bottle (T_s) and cold drink (T_{cd}) and (T_{air}) will remain uniform everywhere and will only change with time (3) The cold drink does not contain any bubble or even if it contain bubbles those will have no effect in heat transfer (4) The time of cooling will be approximated by taking an order of magnitude of the time needed for heat transfer (5) $[(T_{cd} - T_s) = (T_s - T_{air})]$

Find: Whether to place the bottle vertically or horizontally for faster cooling

Solution:

The rate of cooling down of the bottle (Q) will be (assuming no spatial variation of temperature and the cold drink temperature will be uniform and T_{cd} at any time instant),

$$Q = m_{cd} c_{cd} \frac{dT_{cd}}{dt} \dots\dots(1) \quad \text{where } c \text{ is the specific heat}$$

Now this heat is dissipated in the air

So,

$$Q = hA(T_{cd} - T_{air}) \dots\dots(2) \quad \text{where } A \text{ is the total bottle area.}$$

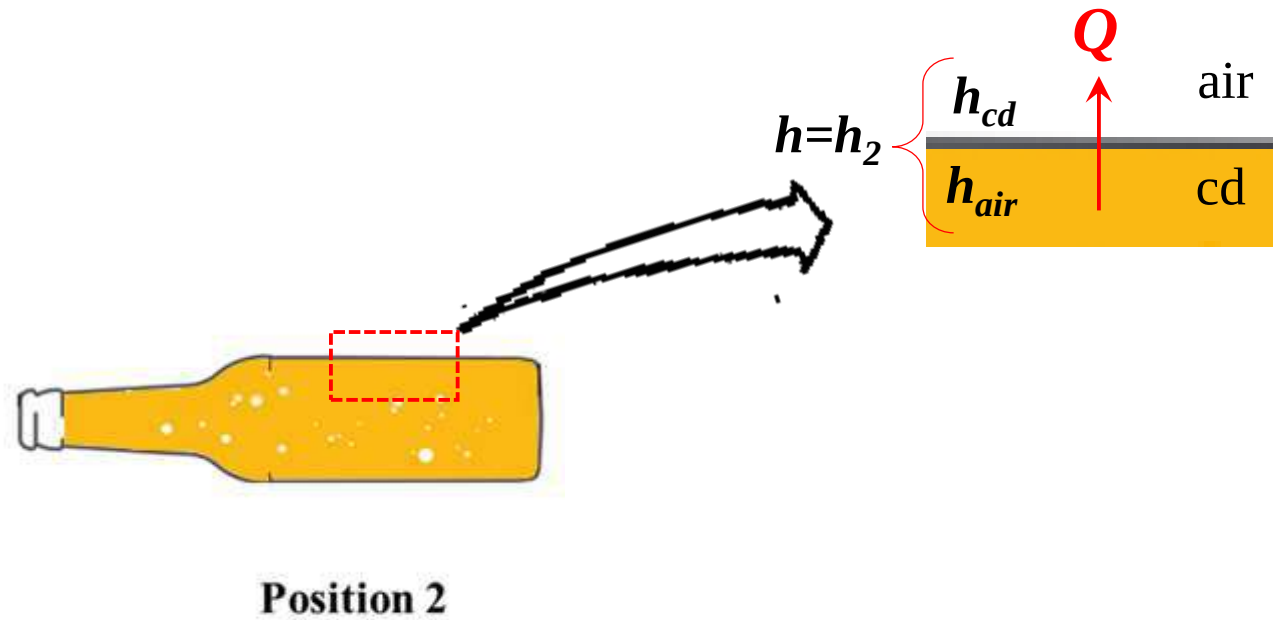
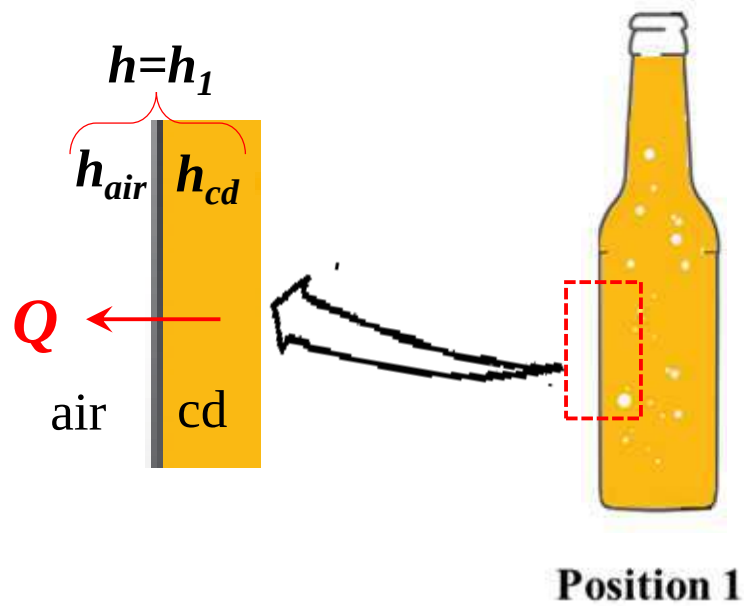
Equating both the equations we will get

$$hA(T_{cd} - T_{air}) = m_{cd}c_{cd} \frac{dT_{cd}}{dt} \quad \dots\dots(3)$$

So, the order of cooling down time for the cold drink bottle will be

$$t \sim \frac{(mc)_{cd}}{hA} \quad \dots\dots(4)$$

If we the above equation carefully only h (heat transfer coefficient) will be a function of the orientation of the bottle. Rest parameters are independent of the orientation.



So, we can write

$$\frac{t_1}{t_2} = \frac{h_2}{h_1} \dots\dots\dots(5)$$

Subscript 1 and 2 stands for vertical and horizontal position respectively

Now we have to estimate the overall heat transfer coefficient (h) for the horizontal and vertical position of the cold drink bottle respectively.

The overall heat transfer coefficient (h) for this problem will be

$$\frac{1}{h} = \frac{1}{h_{cd \text{ side}}} + \frac{1}{h_{air \text{ side}}} \dots\dots\dots(6)$$

Position 1

From the scale analysis for $Pr > 1$ fluids we can write

$$h_{cd} \sim \frac{1}{H} (kRa_H^{1/4})_{cd} \quad \text{and} \quad h_{air} \sim \frac{1}{H} (kRa_H^{1/4})_{air} \dots\dots\dots(7)$$

If we assume the same $\Delta T[(T_{cd} - T_s) = (T_s - T_{air})]$ inside and outside the bottle,

Then we can write

$$\frac{h_{cd}}{h_{air}} \sim \frac{(kRa_H^{1/4})_{cd}}{(kRa_H^{1/4})_{air}} = \frac{\left[k \left(\frac{g\beta\Delta TH^3}{\alpha\nu} \right)^{1/4} \right]_{cd}}{\left[k \left(\frac{g\beta\Delta TH^3}{\alpha\nu} \right)^{1/4} \right]_{air}} = \frac{\left[k \left(\frac{g\beta}{\alpha\nu} \right)^{1/4} \right]_{cd}}{\left[k \left(\frac{g\beta}{\alpha\nu} \right)^{1/4} \right]_{air}} = \frac{0.58 \times (4910)^{1/4}}{0.025 \times (125)^{1/4}} = 58.1$$

Since, $h_{cd} \gg h_{air}$ we can say the overall h is

$$h_1 \sim h_{air} \sim \frac{1}{H} (kRa_H^{1/4})_{air} = \frac{1}{H} \times \left[k \left(\frac{g\beta\Delta TH^3}{\alpha\nu} \right)^{1/4} \right]_{air} = (\text{Constant}) \times H^{-\frac{1}{4}} \dots\dots(8)$$

Similarly,

Position 2

$$h_2 \sim (\text{Constant}) \times L^{-\frac{1}{4}}$$

From equation (5) we can write

$$\frac{t_1}{t_2} = \frac{h_2}{h_1} = \left(\frac{L}{H} \right)^{-1/4} = \left(\frac{L}{5L} \right)^{-1/4} = 5^{1/4} = 1.50$$

So, the cooling the cold drink bottle by placing it vertically will take 50% more time compare to the horizontal position. So, if you want to have the cold drink as soon as possible place it horizontally

Problem 2

The ABC Evening News Report in a news segment on hypothermia research studies at the University of Minnesota claimed that heat loss from the body (at 25°C) is faster in 10°C water than in air at the same temperature. Is that a realistic statement? If the statement is found to be true then how fast? Person can be approximated as a vertical cylinder of diameter $D = 0.3$ m and length $L = 1.8$ m

The variation of Ra with the Nu is given as $Nu_H = 0.1 \times Ra_H^{1/3}$ Where, $Ra_H = \frac{g\beta\Delta TH^3}{\alpha\nu}$

Data:

For air,

$$k = 0.0293 \text{ W/m}\cdot\text{K},$$

$$\nu = 19.91 \times 10^{-6} \text{ m}^2/\text{s},$$

$$\alpha = 28.4 \times 10^{-6} \text{ m}^2/\text{s};$$

$$\beta_a = 3.45 \times 10^{-3} \text{ K}^{-1}$$

For water,

$$k = 0.598 \text{ W/m}\cdot\text{K},$$

$$\nu = 1.081 \times 10^{-6} \text{ m}^2/\text{s},$$

$$\alpha = 1.431 \times 10^{-7} \text{ m}^2/\text{s},$$

$$\beta_w = 174 \times 10^{-6} \text{ K}^{-1}.$$

Solution:

Known: Person, approximated as a cylinder, experiencing heat loss in water or air at 10°C.

Find: Whether heat loss from body in water is faster than that in air. If so then how fast?

Assumptions: (1) Person can be approximated as a vertical cylinder of diameter $D = 0.3$ m and height $H = 1.8$ m (2) Initially the outer surface of the body is kept at 25°C (3) Loss is only from the lateral surface of the body.

Analysis: In both water (w) and air (a), the heat loss from the lateral surface of the cylinder approximating the body is

$$q = h \times \pi D L \times (T_s - T_\infty) \quad \dots\dots(9)$$

here T_s and T_∞ are the same for both situations. Hence,

$$\frac{q_w}{q_a} = \frac{h_w}{h_a} \quad \dots\dots(10)$$

For Vertical cylinder in air:

$$Ra_H = \frac{g\beta\Delta TH^3}{\alpha\nu} = \frac{9.8 \frac{m}{s^2} \times 3.45 \times 10^{-3} \frac{1}{K} \times (25-10)K \times (1.8m)^3}{19.91 \times 10^{-6} \frac{m^2}{s} \times 28.4 \times 10^{-6} \frac{m^2}{s}} = 5.228 \times 10^9$$

So, from the relation with Nu

$$Nu_H = 0.1 \times Ra_H^{1/3} = 0.1 \times (5.228 \times 10^9)^{1/3} = 173.4$$

$$Nu_H = \frac{h \times H}{k} = 173.4 \quad \Rightarrow \quad h = 2.82 \text{ W / m}^2 \text{ K}$$

Vertical cylinder in water:

$$Ra_H = \frac{g \beta \Delta T H^3}{\alpha \nu} = \frac{9.8 \frac{\text{m}}{\text{s}^2} \times 174 \times 10^{-6} \frac{1}{\text{K}} \times (25 - 10) \text{ K} \times (1.8 \text{ m})^3}{1.081 \times 10^{-6} \frac{\text{m}^2}{\text{s}} \times 1.431 \times 10^{-7} \frac{\text{m}^2}{\text{s}}} = 9.643 \times 10^{11}$$

So, from the relation with Nu

$$Nu_H = 0.1 \times Ra_H^{1/3} = 0.1 \times (9.643 \times 10^{11})^{1/3} = 978.9$$

$$Nu_H = \frac{h \times H}{k} = 978.9 \quad \Rightarrow \quad h = 328 \text{ W} / \text{m}^2 \text{K}$$

Hence, from this analysis we find (substituting in equation 10)

$$\frac{q_w}{q_a} = \frac{328 \text{ W} / \text{m}^2 \text{K}}{2.8 \text{ W} / \text{m}^2 \text{K}} = 117 \quad \Rightarrow \quad q_w = 117 \times q_a$$

Conclusion:

Heat loss from the body is 117 times faster in 10°C water than in air at the same temperature.

Summary of flow and heat transfer scales in a natural convection boundary layer along a vertical wall

Prandtl Number Range	Thermal Boundary Layer Thickness	Wall Jet Velocity Profile			Nusselt Number $Nu = \frac{hH}{k}$
		Distance from Wall to Velocity Peak	Thickness of Wall Jet	Velocity Scale	
$Pr > 1$	$H Ra_H^{-1/4}$	$H Ra_H^{-1/4}$	$Pr^{1/2} (H Ra_H^{-1/4})$	$\frac{\alpha}{H} Ra_H^{1/2}$	$Ra_H^{1/4}$
$Pr < 1$	$Pr^{-1/4} (H Ra_H^{-1/4})$	$Pr^{1/4} (H Ra_H^{-1/4})$	$Pr^{-1/4} (H Ra_H^{-1/4})$	$\frac{\alpha}{H} (Pr Ra_H)^{1/2}$	$(Pr Ra_H)^{1/4}$

Integral solutions

Integrating the momentum equation and the energy equation from the wall ($x = 0$) to a far enough plane $x = X$ in the motionless isothermal cold reservoir, we obtain the integral boundary layer equations for momentum and energy,

$$\frac{d}{dy} \int_0^X v^2 dx = -v \left(\frac{\partial v}{\partial x} \right)_{x=0} + g\beta \int_0^X (T - T_\infty) dx \quad (1)$$

$$\frac{d}{dy} \int_0^X v(T_\infty - T) dx = \alpha \left(\frac{\partial T}{\partial x} \right)_{x=0} \quad (2)$$

High-Pr Fluids

A suitable set of profiles for $\text{Pr} > 1$ fluids

$$\begin{aligned} T - T_\infty &= \Delta T e^{-x/\delta_T} \\ v &= V e^{-x/\delta} (1 - e^{-x/\delta_T}) \end{aligned} \quad (3)$$

1. In the limit $\text{Pr} \rightarrow \infty$, this solution reduces to

$$\frac{\delta}{\delta_T} = \left(\frac{6}{5} \text{Pr} \right)^{1/2} \quad \text{and} \quad \text{Nu} = \mathbf{0.783} \text{Ra}_y^{1/4}$$

Low-Pr Fluids

For a $\text{Pr} < 1$ fluid, the temperature profile is combined with a new velocity profile:

$$v = V_1 e^{-x/\delta_T} (1 - e^{-x/\delta_v}) \quad (4)$$

In the limit $\text{Pr} \rightarrow 0$, these results become

$$\frac{\delta_v}{\delta_T} = \left(\frac{3}{5} \text{Pr} \right)^{1/2} \quad \text{and} \quad \text{Nu} = \mathbf{0.689} (\text{Pr Ra}_y)^{1/4}$$

Similarity solutions

➤ The similarity variable emerges as

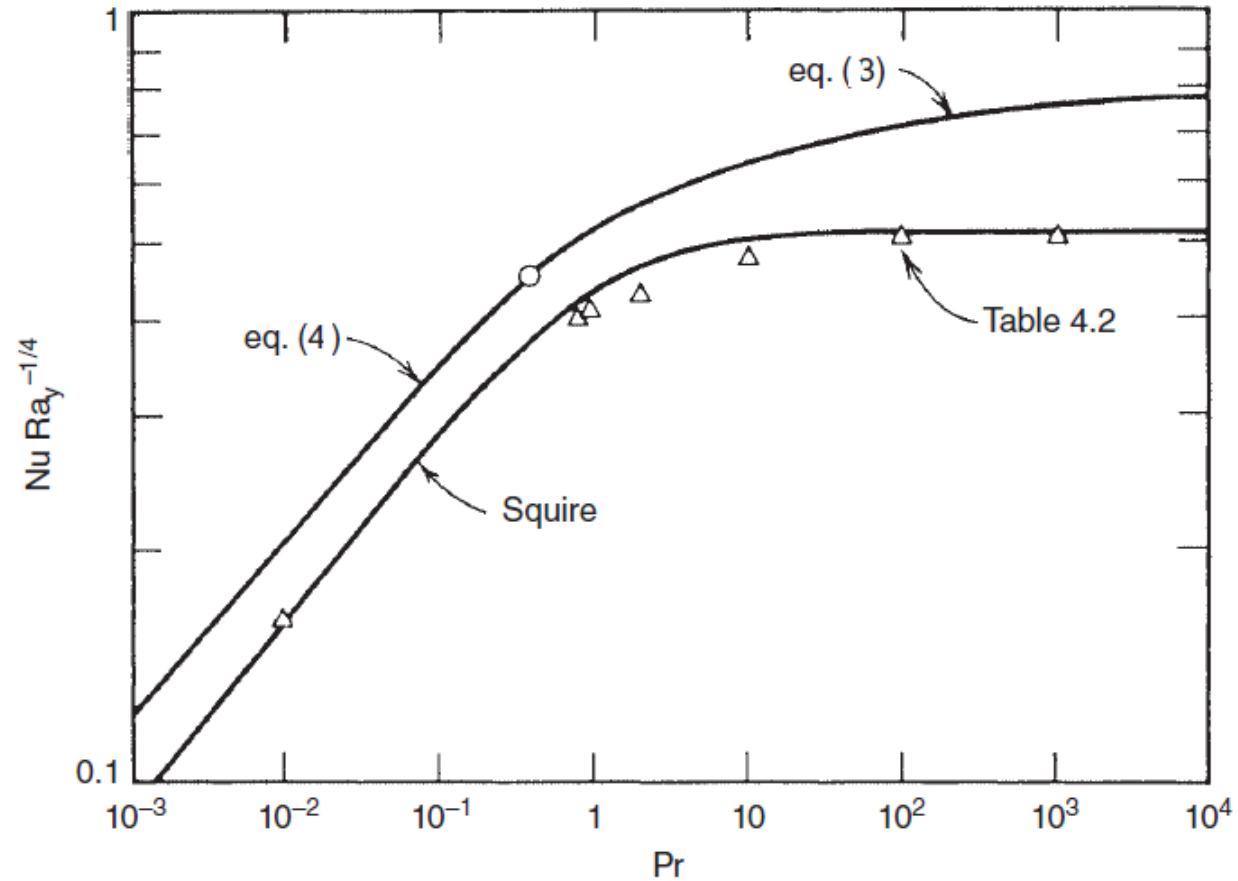
$$\eta = \frac{x}{y} \text{Ra}_y^{1/4} \quad \text{and} \quad \frac{T - T_\infty}{T_0 - T_\infty} = \theta(\eta, \text{Pr})$$

Table 4.2 Similarity solution heat transfer results for natural convection boundary layer along a vertical isothermal wall^a

Pr	0.01	0.72	1	2	10	100	1000
Nu Ra _y ^{-1/4}	0.162	0.387	0.401	0.426	0.465	0.490	0.499

$$\text{Nu} = \mathbf{0.503} \text{Ra}_y^{1/4} \quad \text{as } \text{Pr} \rightarrow \infty$$

$$\text{Nu} = \mathbf{0.6}(\text{Pr Ra}_y)^{1/4} \quad \text{as } \text{Pr} \rightarrow 0$$



H. B. Squire, in S. Goldstein, ed., *Modern Developments in Fluid Dynamics*, Vol. II, Dover, New York, 1965, pp. 641–643.

Table 4.2 Similarity solution heat transfer results for natural convection boundary layer along a vertical isothermal wall^a

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$Nu Ra_y^{-1/4}$	0.162	0.387	0.401	0.426	0.465	0.490	0.499

^aNumerical values calculated from Ostrach's solution [16].

HEAT TRANSFER RESULTS INCLUDING THE EFFECT OF TURBULENCE

Vertical Walls

Transition to turbulent flow occurs at the y position where

$$\text{Gr}_y \sim 10^9 \quad (10^{-3} \leq \text{Pr} \leq 10^3)$$

$$\text{Ra}_y \sim 10^9 \text{Pr} \quad (10^{-3} \leq \text{Pr} \leq 10^3)$$

An empirical isothermal-wall correlation that reports the wall-averaged Nusselt number Nu_y for the entire Rayleigh number range—laminar, transition, turbulent

$$\overline{\text{Nu}}_y = \left\{ 0.825 + \frac{0.387 \text{Ra}_y^{1/6}}{[1 + (0.492 / \text{Pr})^{9/16}]^{8/27}} \right\}^2$$

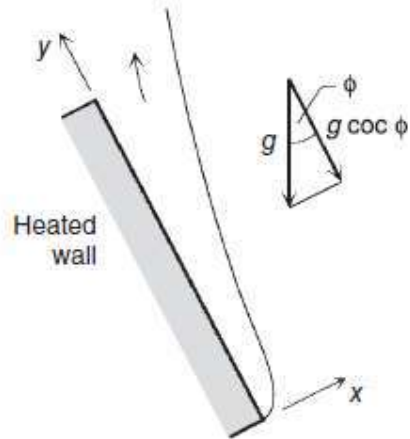
Churchill and Chu [33]

When the vertical wall is heated uniformly

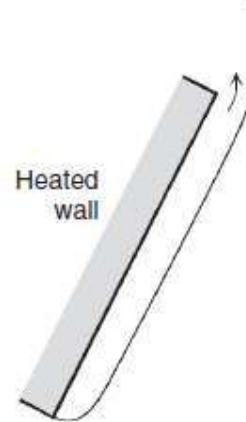
$$\left. \begin{array}{l} \text{Nu}_y = 0.6\text{Ra}_{*y}^{1/5} \\ \overline{\text{Nu}}_y = 0.75\text{Ra}_{*y}^{1/5} \end{array} \right\} \begin{array}{l} \text{laminar,} \\ 10^5 < \text{Ra}_{*y} < 10^{13} \end{array}$$

$$\left. \begin{array}{l} \text{Nu}_y = 0.568\text{Ra}_{*y}^{0.22} \\ \overline{\text{Nu}}_y = 0.645\text{Ra}_{*y}^{0.22} \end{array} \right\} \begin{array}{l} \text{turbulent,} \\ 10^{13} < \text{Ra}_{*y} < 10^{16} \end{array}$$

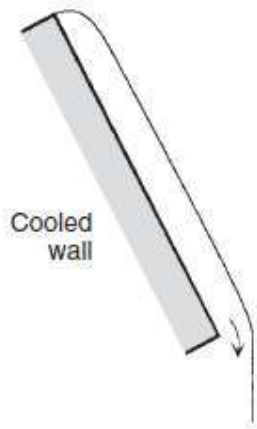
Inclined Walls



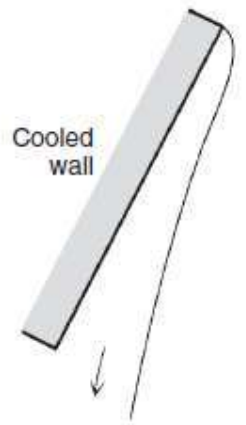
(a)



(b)



(c)



(d)

Isothermal wall

$$Ra_y = \frac{g \cos \phi \beta (T_w - T_\infty) y^3}{\alpha \nu}$$

Uniform heat flux

$$Ra_{*y} = \frac{g \cos \phi \beta q''_w y^4}{\alpha \nu k}$$

ϕ is restricted to the range $-60^\circ < \phi < 60^\circ$

In the ***laminar regime*** momentum equation is analogous, except that $g \cos \phi$ replaces g in the buoyancy term.

laminar–turbulent transition,

The uniform-flux wall is oriented as in cases *a* and *d*

ϕ	Ra_{*y}
0°	$5 \times 10^{12} - 10^{14}$
30°	$3 \times 10^{10} - 10^{12}$
60°	$6 \times 10^7 - 6 \times 10^9$

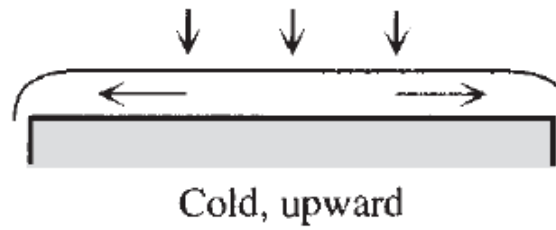
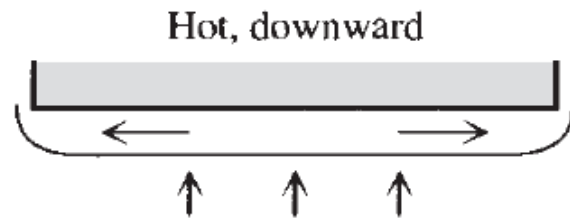
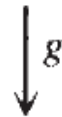
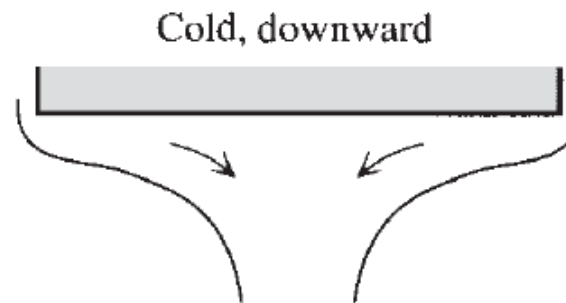
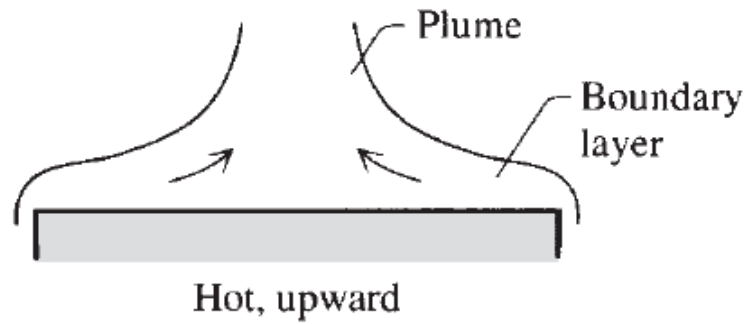
in water experiments ($Pr \sim 6.5$)

laminar–turbulent transition

Effect of wall inclination on transition along an *isothermal* wall [37]

ϕ	Ra_y
0°	8.7×10^8
20°	2.5×10^8
45°	1.7×10^7
60°	7.7×10^5

Horizontal Walls



Characteristic length

$$L = \frac{A}{p}$$

A is the area of the plane surface
 p is the perimeter

Isothermal surfaces

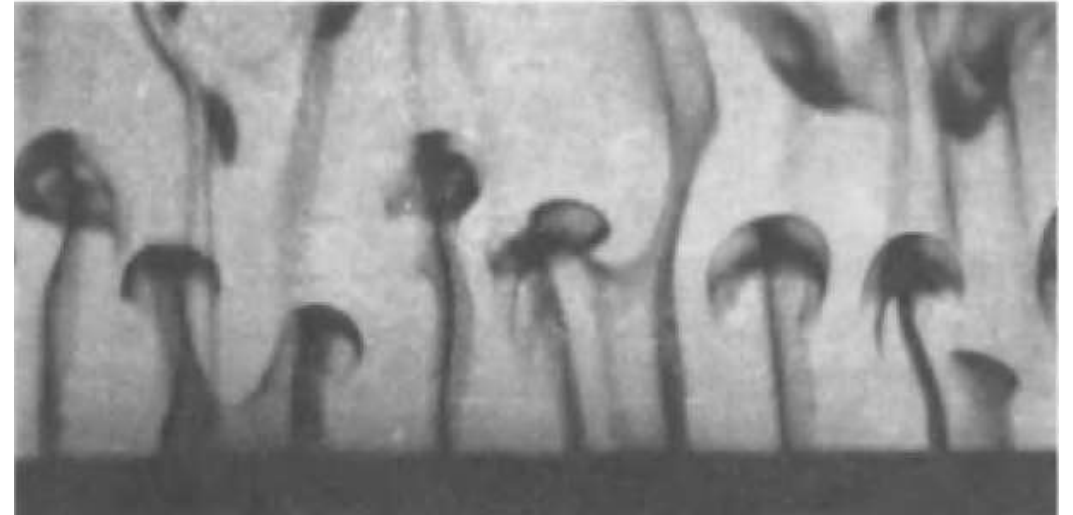
In the case of hot surfaces facing upward or cold surfaces facing downward the Nusselt number varies as

$$\overline{\text{Nu}}_L = \begin{cases} 0.54\text{Ra}_L^{1/4} & (10^4 < \text{Ra}_L < 10^7) \\ 0.15\text{Ra}_L^{1/3} & (10^7 < \text{Ra}_L < 10^9) \end{cases} \quad \text{Pr} > 0.5$$

Correlation for hot surfaces facing downward or cold surfaces facing upward (Fig. bottom) is

$$\overline{\text{Nu}}_L = 0.27\text{Ra}_L^{1/4} \quad (10^5 < \text{Ra}_L < 10^{10})$$

Thermals

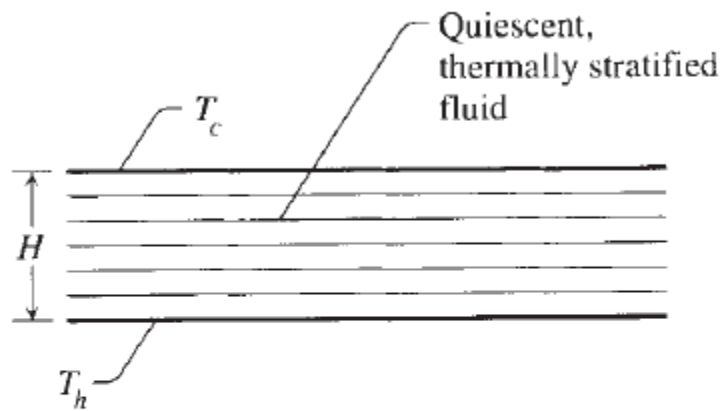


Intermittent rise of heated fluid (thermals) from a horizontal surface facing upward in a colder pool of water *Journal of Fluid Mechanics*, Vol. 41, pp. 793–800, 1970.

When the temperature difference is sufficiently large

Internal Natural Convection

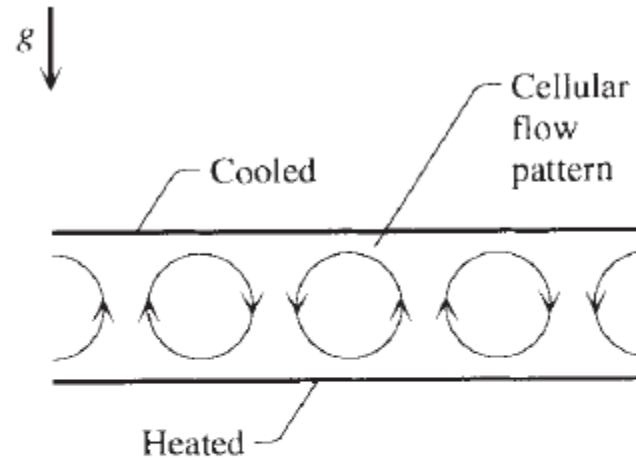
- (1) enclosures heated from the side.
- (2) enclosures heated from below.



$$Ra_H < 1708$$

$$\overline{Nu}_H = 1$$

(a)



$$Ra_H > 1708$$

$$\overline{Nu}_H > 1$$

(b)



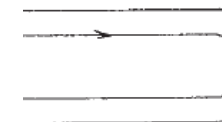
Regime II



Regime III

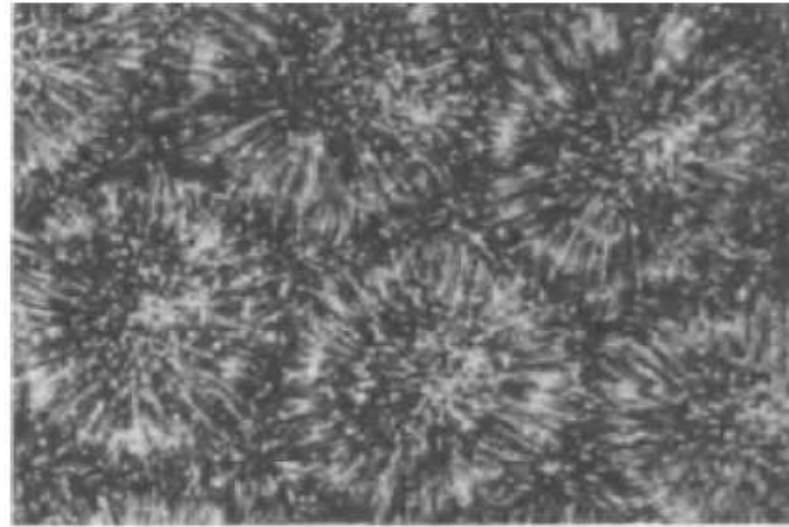
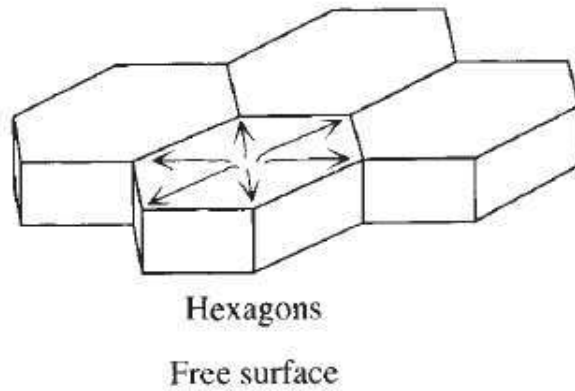
Critical Rayleigh number

$$Ra_H > \sim 1708$$



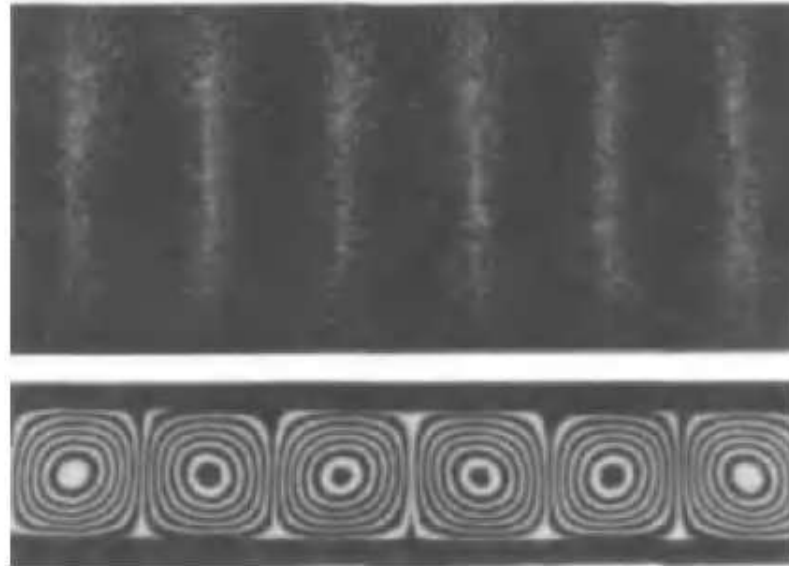
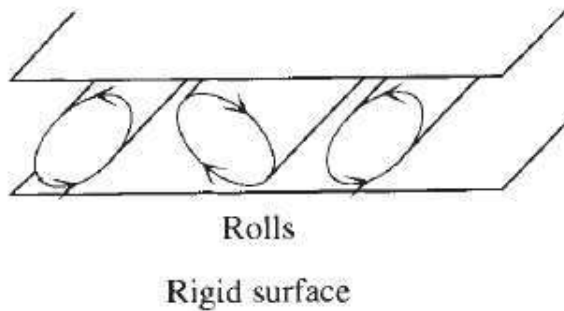
Regime IV

on patterns associated



Rayleigh–Bénard convection in soup

<https://www.youtube.com/watch?v=nQUH9nGTZTY>



Rayleigh-Benard convection in glucose syrup

<https://www.youtube.com/watch?v=b-XrNPqyC48>

Figure 5.21 Rolls and hexagonal cells in a fluid layer heated from below. (Reprinted with permission from J. Zierp and H. Oertel, Jr., eds., *Convective Transport and Instability Phenomena*, 1982, p. 10. Copyright © 1982 G. Braun, Karlsruhe, Germany).

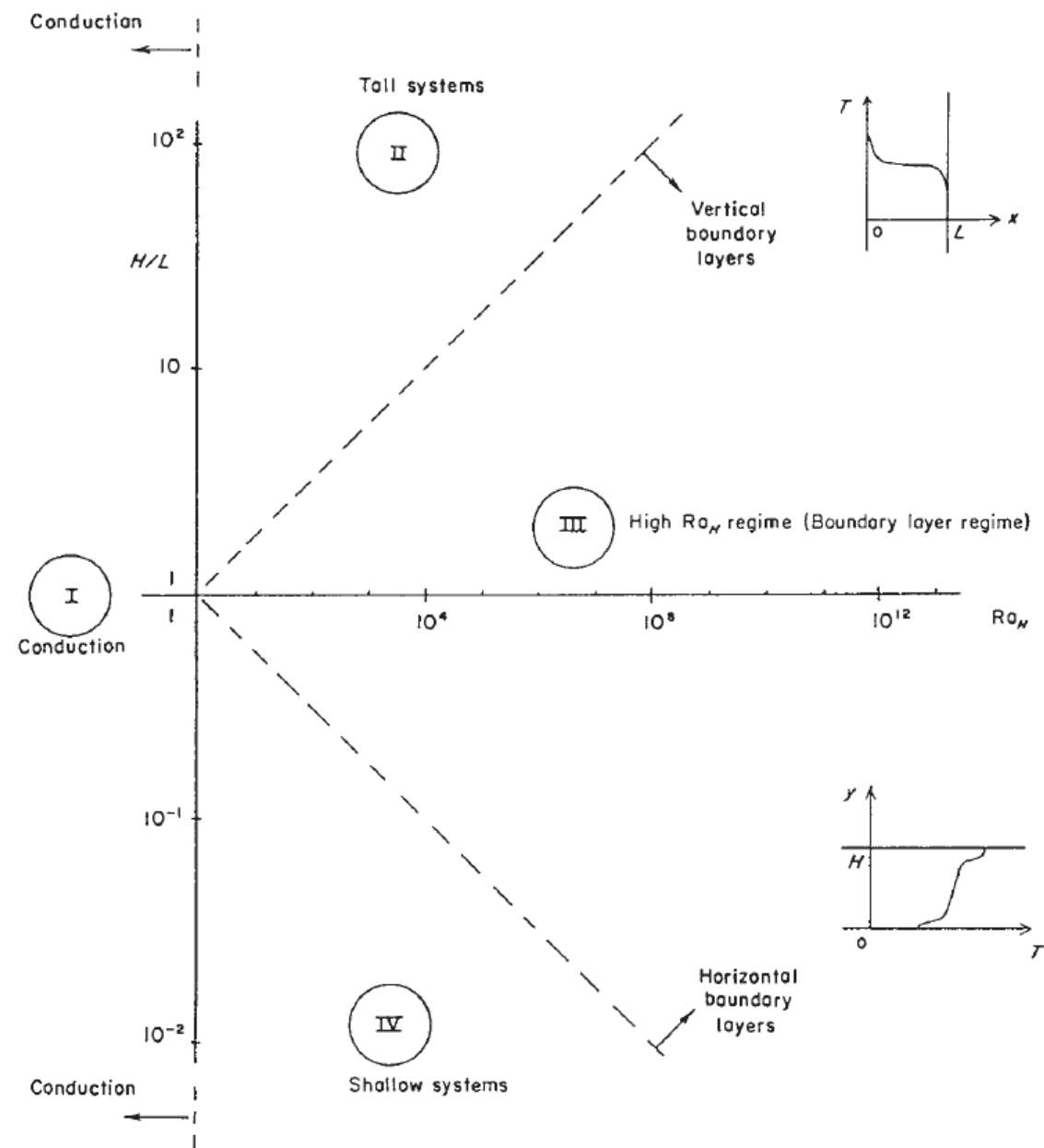


Figure 5.3 Four heat transfer regimes for natural convection in an enclosure heated from the side.